

Bitwise linear trail through **LBlock** found by SAT

CiVerLy

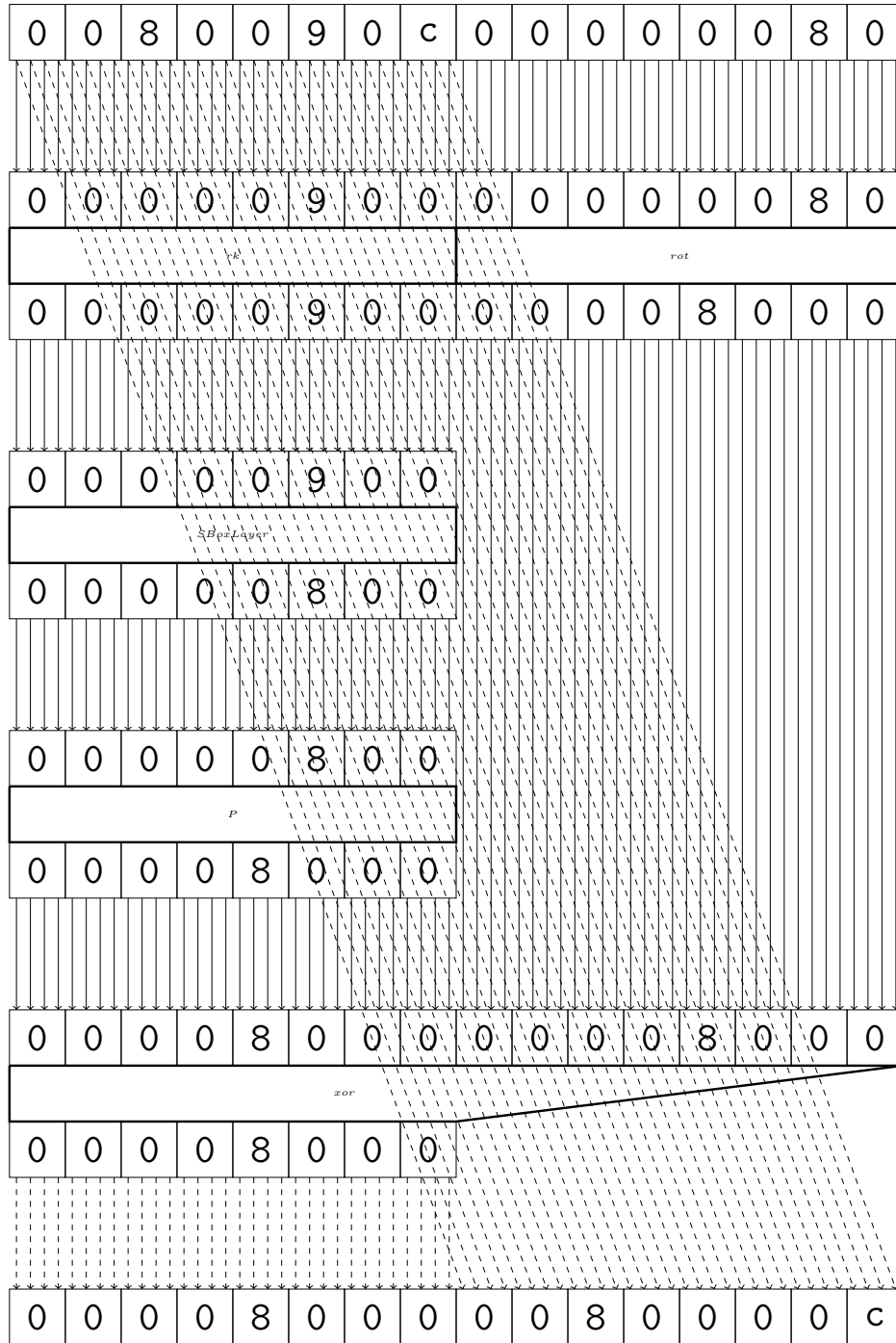
July 20, 2026

1 LBlock

Maximal linear correlation: 2^{-40}

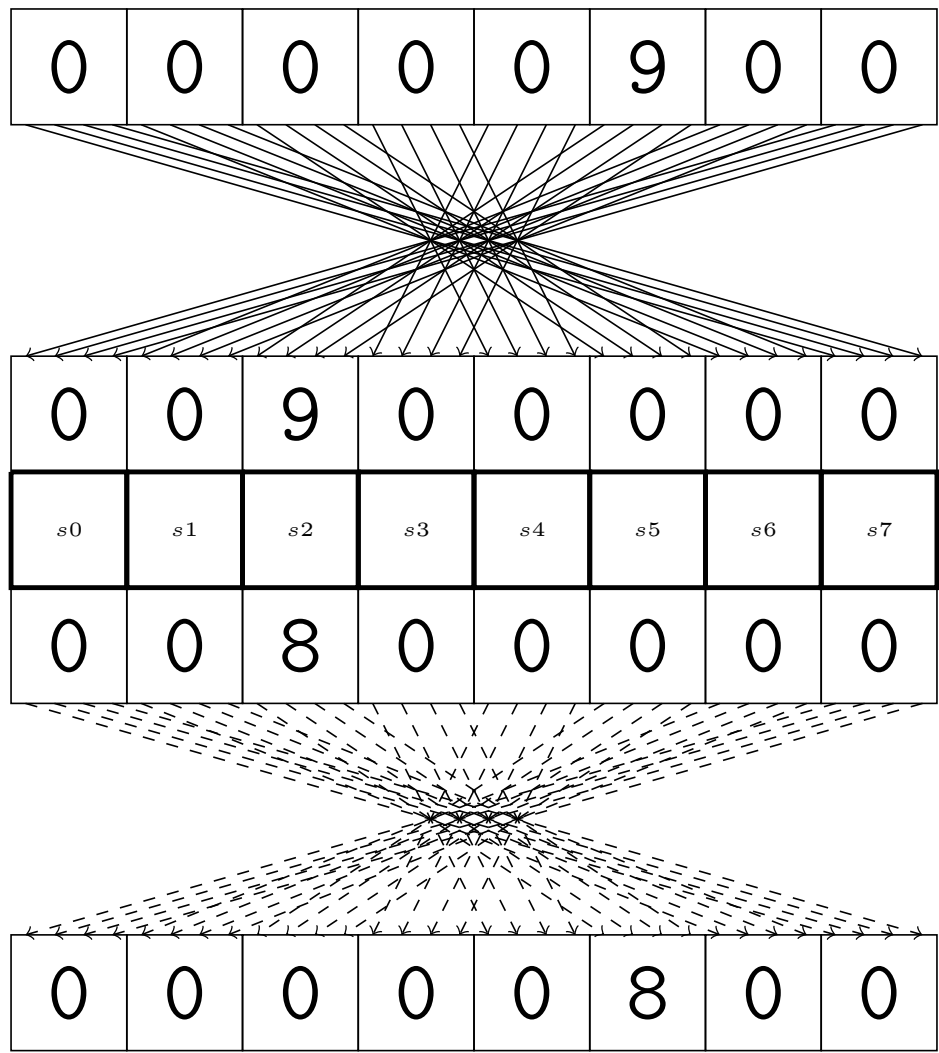
0	0	8	0	0	9	0	c	0	0	0	0	0	8	0		
0	0	8	0	0	9	0	c	0	0	0	0	0	0	8	0	
0	0	0	0	8	0	0	0	0	0	0	8	0	0	0	c	
0	0	0	0	8	0	0	0	0	0	0	8	0	0	0	c	
8	0	0	0	c	0	0	0	8	0	0	8	0	0	e		
8	0	0	0	c	0	0	0	8	0	0	8	0	0	e	*	
0	0	8	0	0	e	0	8	0	0	0	0	0	0	c	8	a
0	0	8	0	0	e	0	8	0	0	0	0	0	0	c	8	a
0	0	0	c	8	a	0	0	0	0	0	0	0	0	0	0	0
0	0	0	c	8	a	0	0	0	0	0	0	0	0	0	0	0
0	0	0	0	0	0	0	0	0	0	0	c	8	a	0	0	
0	0	0	0	0	0	0	0	0	0	0	c	8	a	0	0	
0	c	8	a	0	0	0	0	8	0	e	8	0	0	0	0	
0	c	8	a	0	0	0	0	8	0	e	8	0	0	0	0	
e	8	0	0	0	8	0	0	a	8	0	c	0	0	0	0	
e	8	0	0	0	8	0	0	a	8	0	c	0	0	0	0	
8	0	c	0	0	0	a	0	0	0	0	0	0	0	0	0	
8	0	c	0	0	0	a	0	0	0	0	0	0	0	0	0	
0	0	0	0	0	0	0	0	8	0	c	0	0	0	0	a	
0	0	0	0	0	0	0	0	8	0	c	0	0	0	0	a	
c	0	0	0	a	8	0	0	e	0	0	4	0	0	8		
c	0	0	0	a	8	0	0	e	0	0	4	0	0	8		
0	0	4	0	0	8	0	e	0	0	0	0	a	e	8		
0	0	4	0	0	8	0	e	0	0	0	0	a	e	8		
0	0	0	a	e	8	0	0	0	0	0	0	0	0	0	0	
0	0	0	a	e	8	0	0	0	0	0	0	0	0	0	0	
0	0	0	0	0	0	0	0	0	0	a	e	8	0	0		
0	0	0	0	0	0	0	0	0	0	a	e	8	0	0		
0	a	e	8	0	0	0	6	0	1	c	0	0	0	0		
0	a	e	8	0	0	0	6	0	1	c	0	0	0	0		
1	c	0	0	0	6	0	0	3	e	0	8	0	0	0		
1	c	0	0	0	6	0	0	3	e	0	8	0	0	0		
e	0	8	0	0	0	3	0	0	0	0	0	0	0	0	0	
e	0	8	0	0	0	3	0	0	0	0	0	0	0	0	0	
0	0	0	0	0	0	0	e	0	8	0	0	0	0	3		
0	0	0	0	0	0	0	e	0	8	0	0	0	0	3		
8	0	0	0	0	3	e	0	0	8	0	0	4	0	0	2	
8	0	0	0	0	3	e	0	0	8	0	0	4	0	0	2	
0	8	0	0	4	0	0	2	8	0	0	0	0	3	e	0	
0	8	0	0	4	0	0	2	8	0	0	0	0	3	e	0	

2 LBlock_round

Maximal linear correlation: $2^{-1.0}$ 

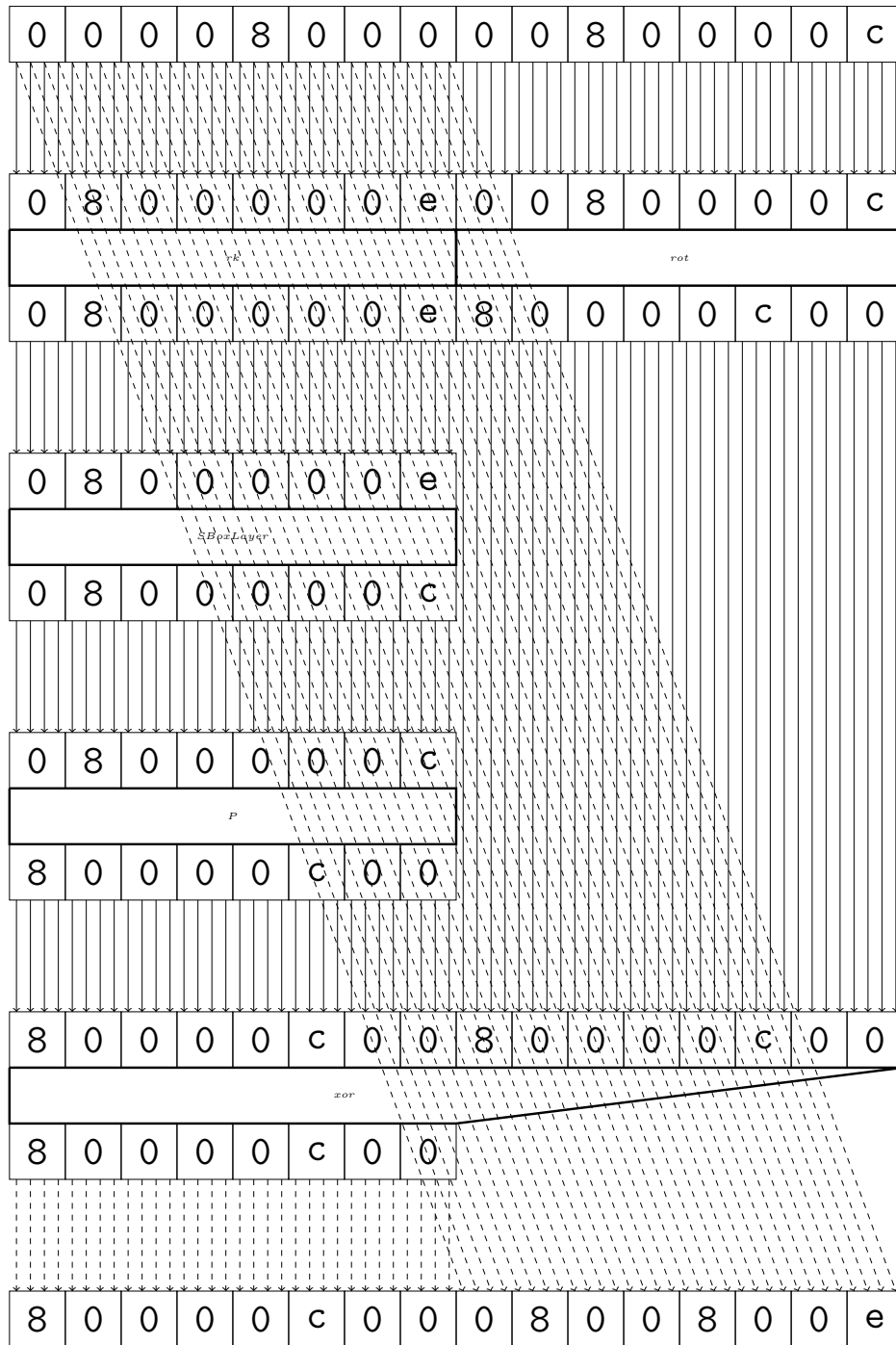
3 SBoxLayer

Maximal linear correlation: $2^{-1.0}$



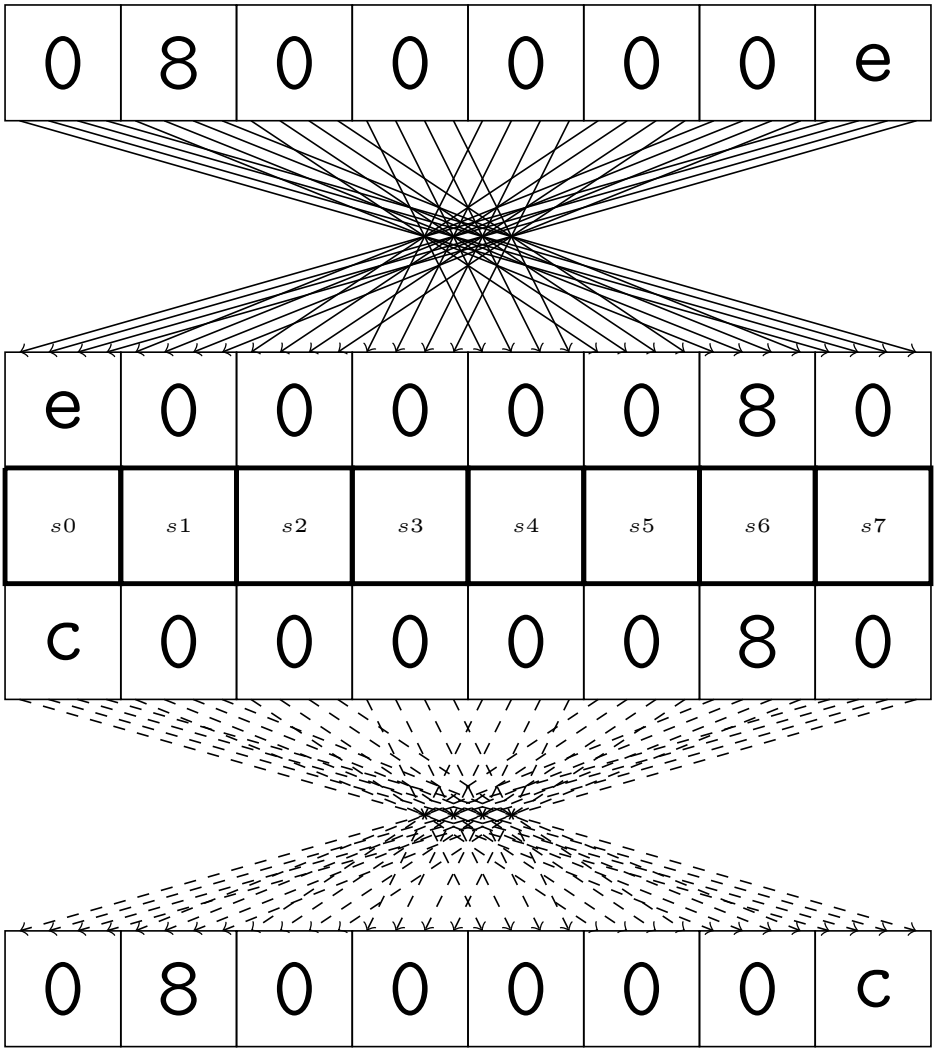
4 LBlock_round

Maximal linear correlation: $2^{-2.0}$

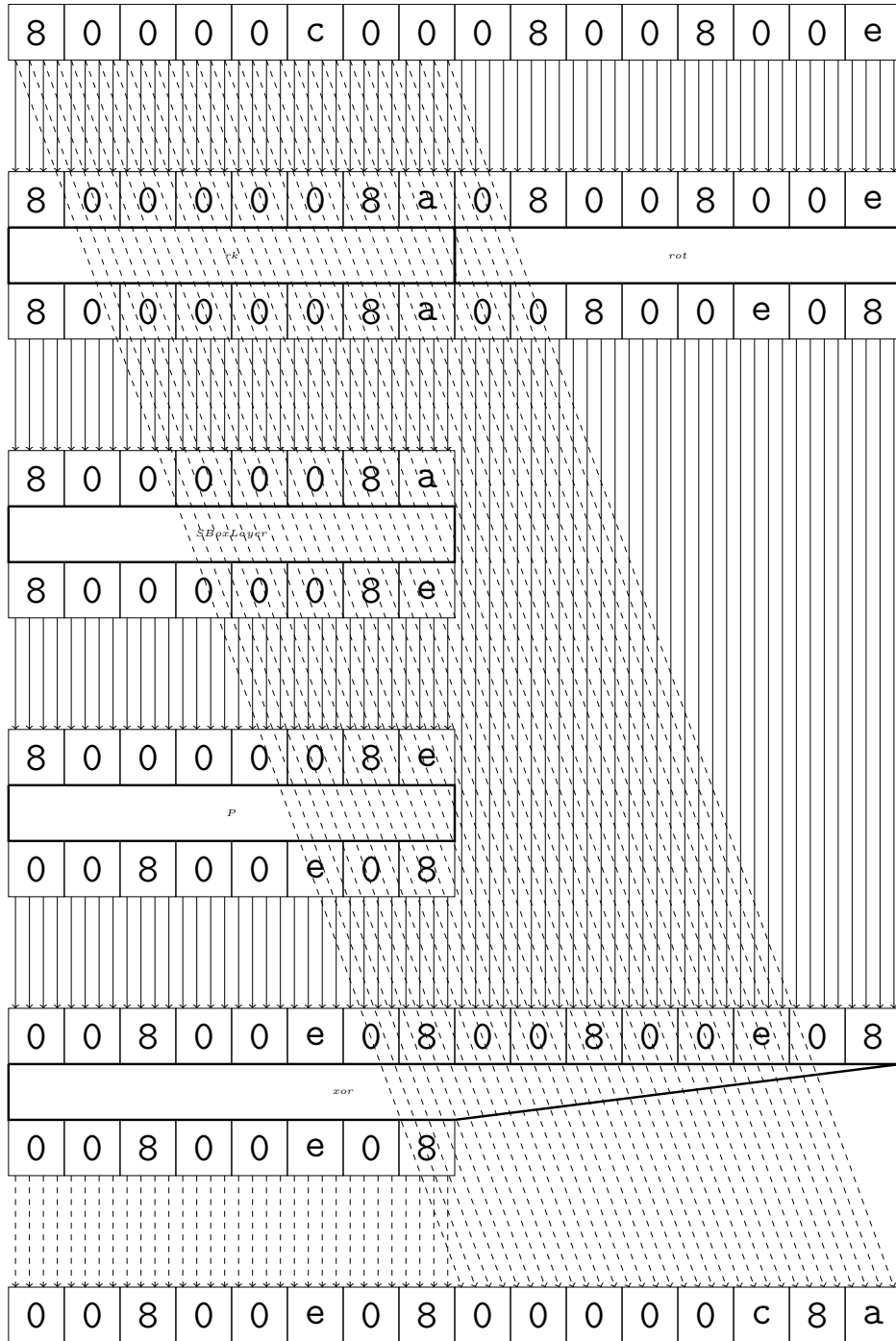


5 SBoxLayer

Maximal linear correlation: $2^{-2.0}$

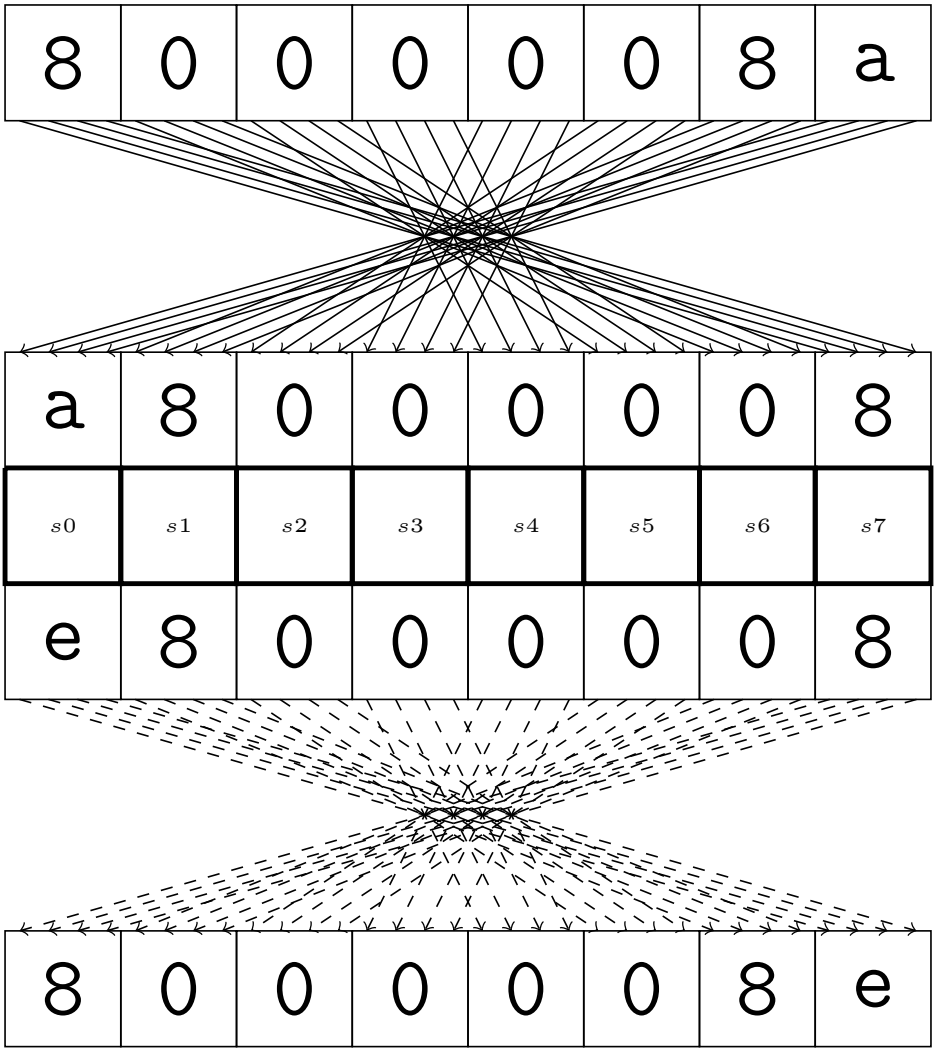


6 LBlock_round

Maximal linear correlation: $2^{-3.0}$ 

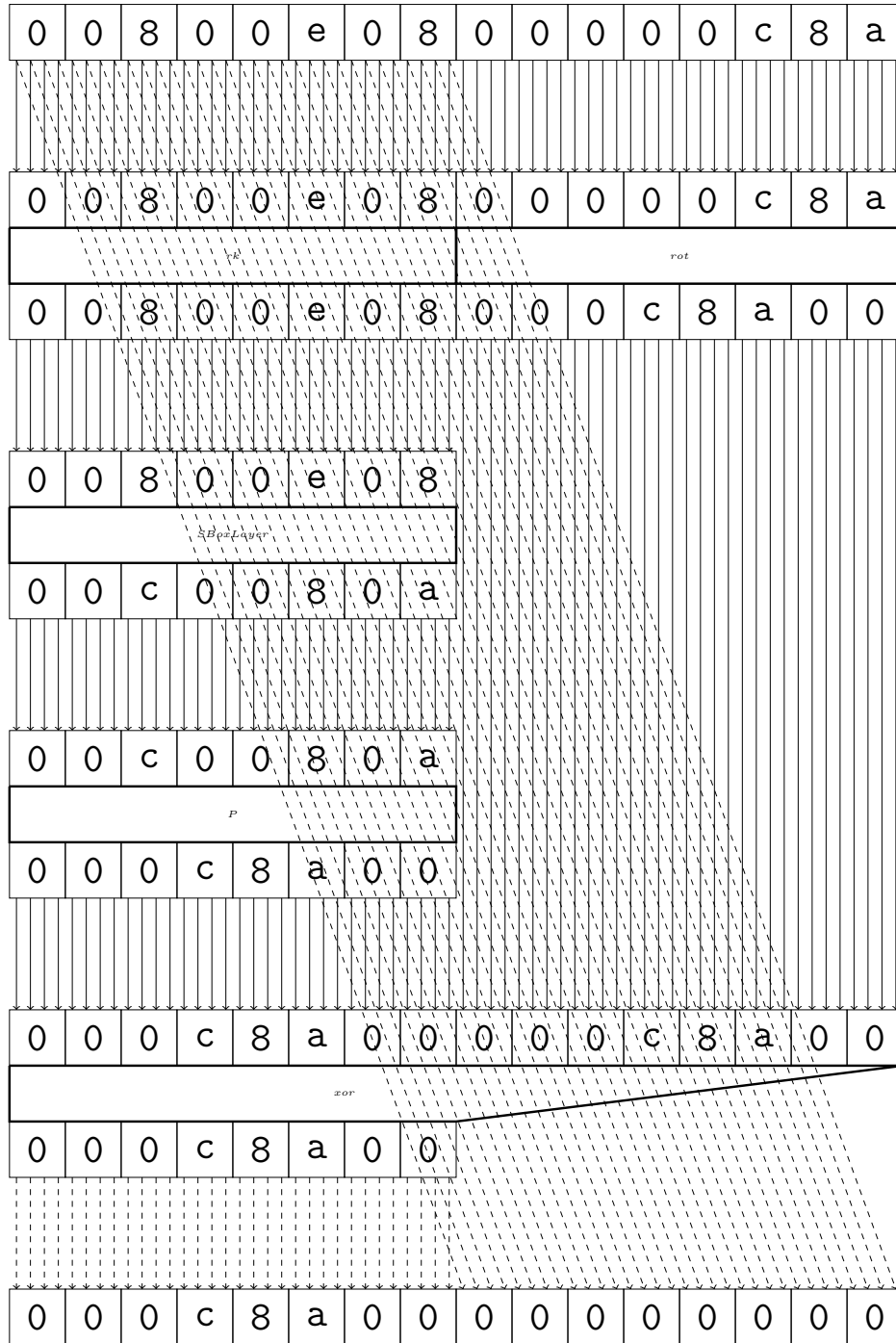
7 SBoxLayer

Maximal linear correlation: $2^{-3.0}$



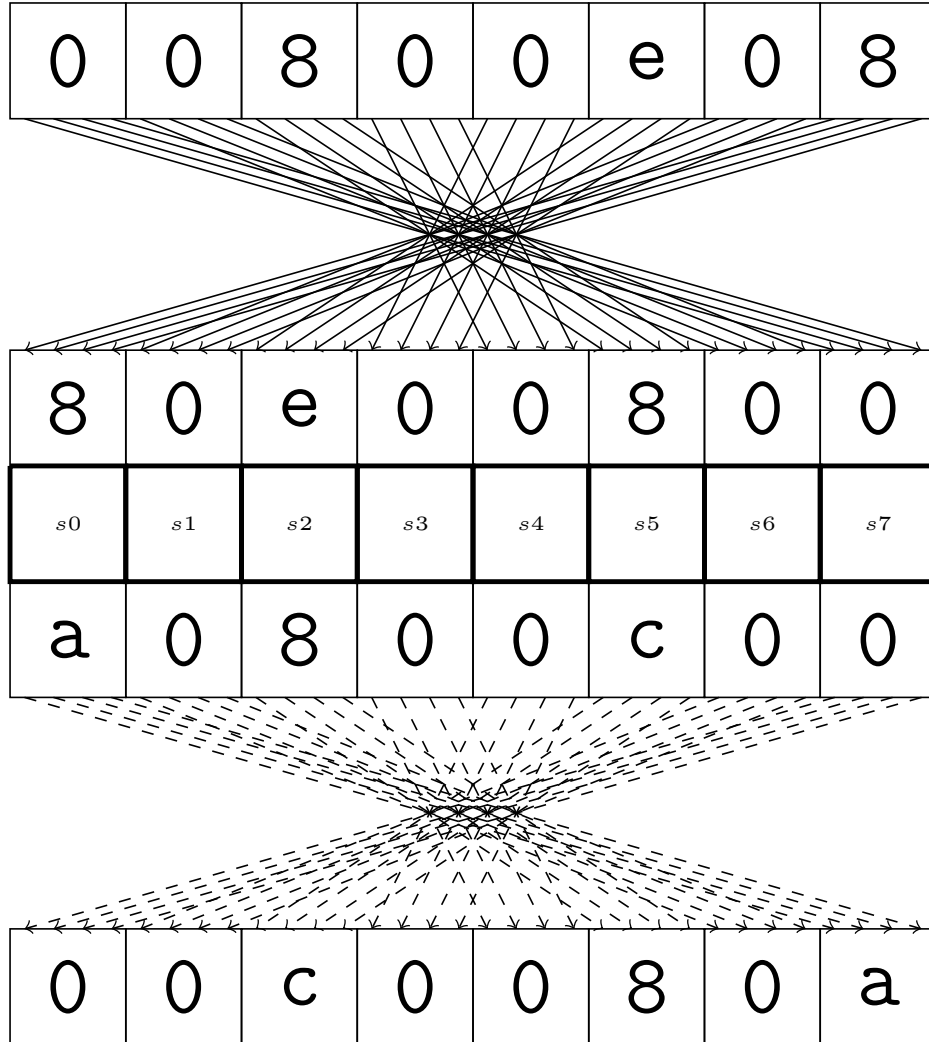
8 LBlock_round

Maximal linear correlation: $2^{-3.0}$



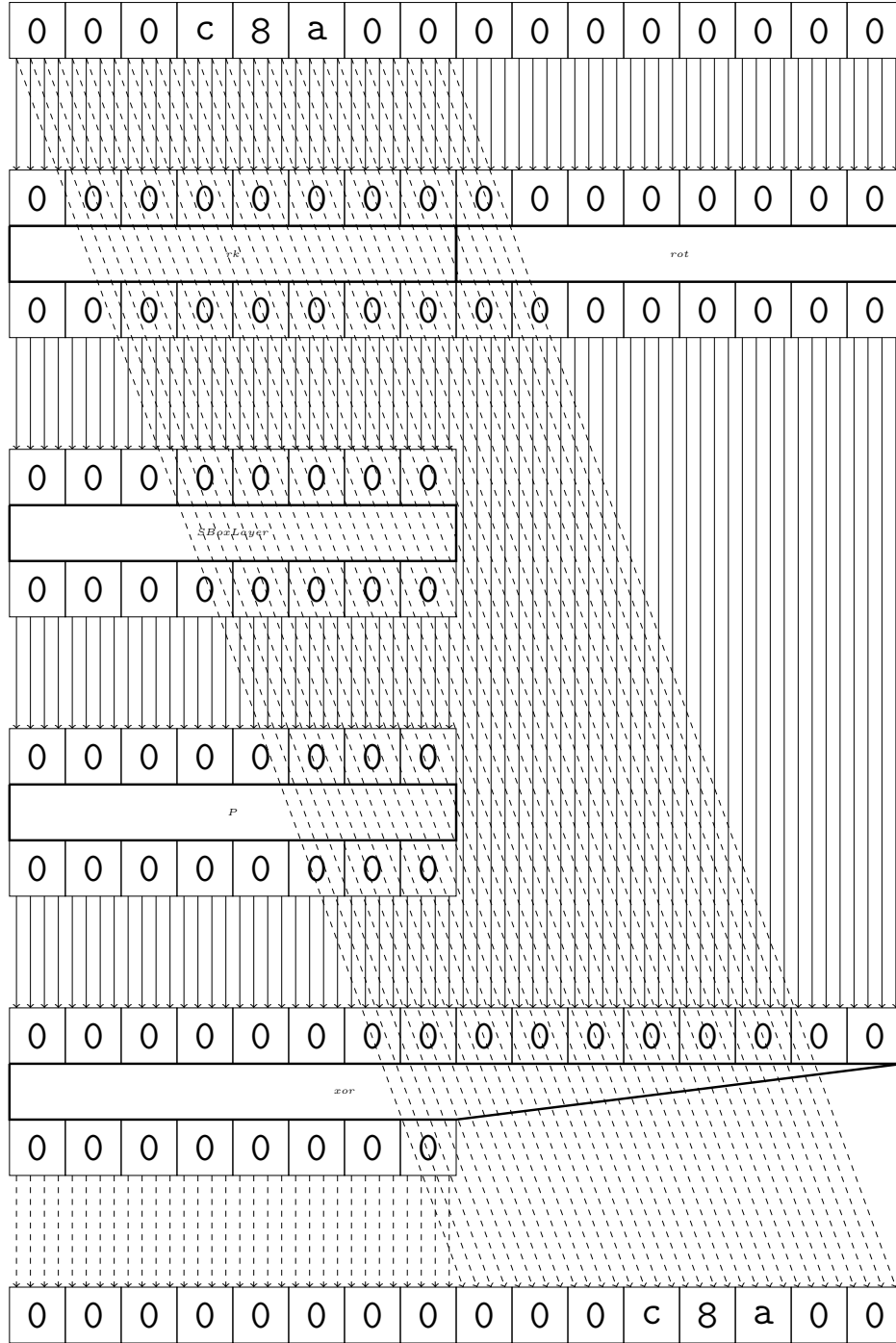
9 SBoxLayer

Maximal linear correlation: $2^{-3.0}$



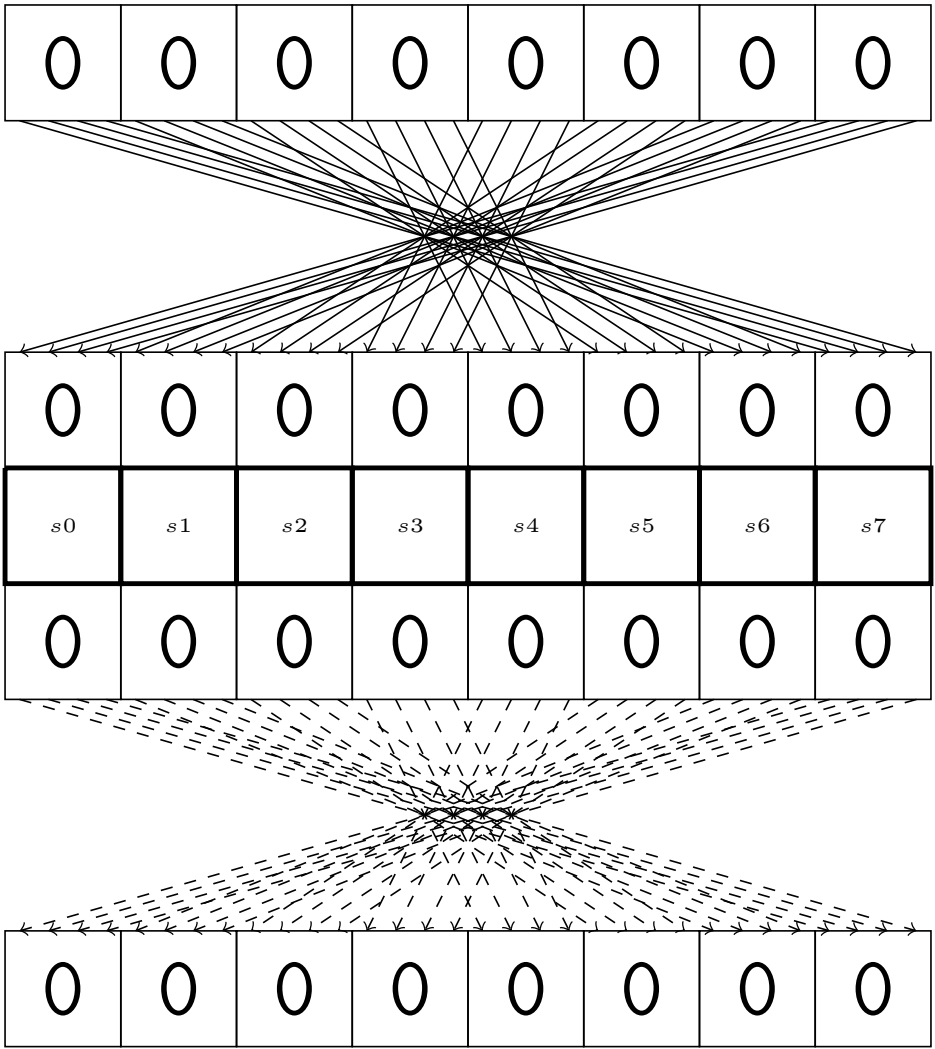
10 LBlock_round

Maximal linear correlation: $2^{-0.0}$

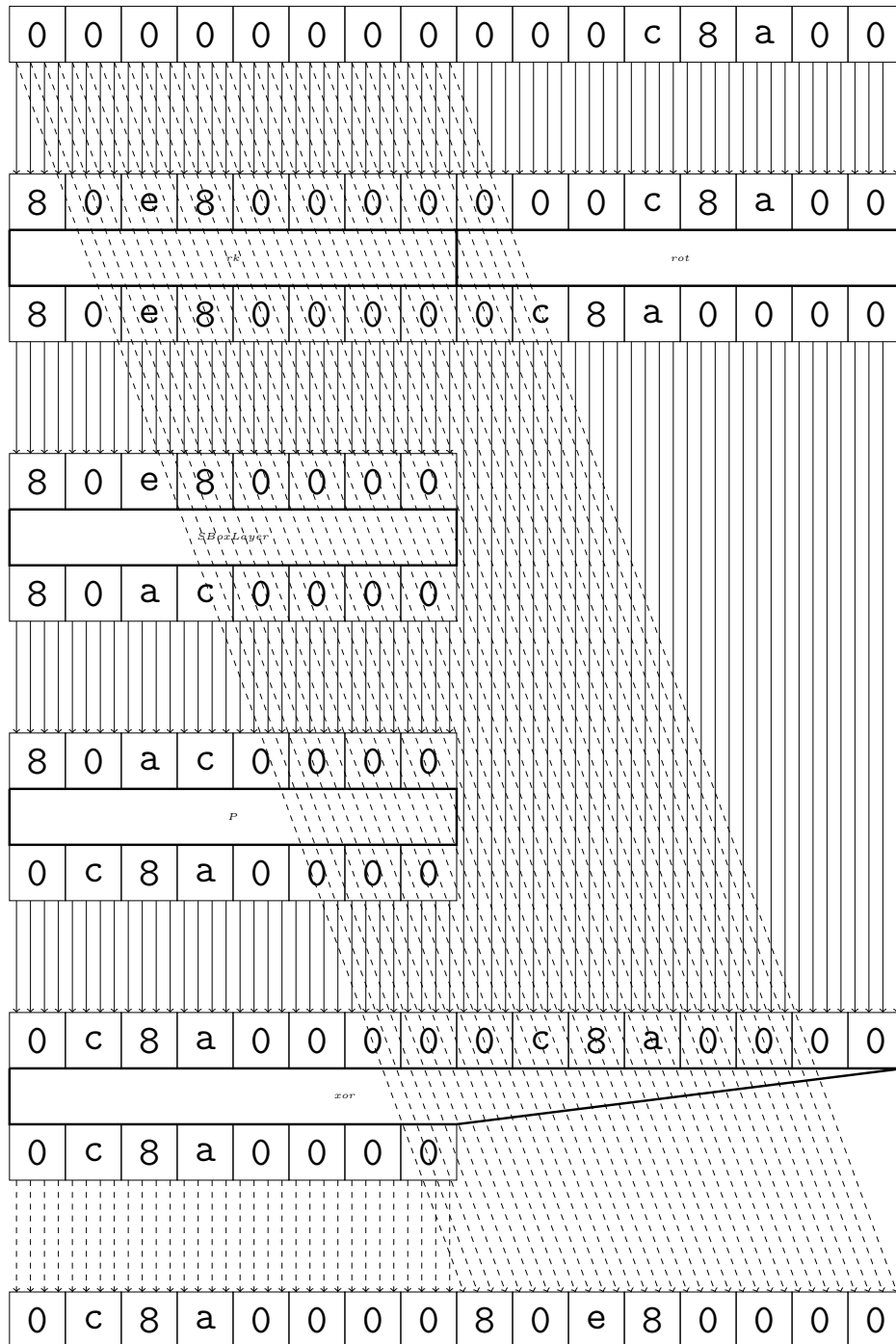


11 SBoxLayer

Maximal linear correlation: $2^{-0.0}$

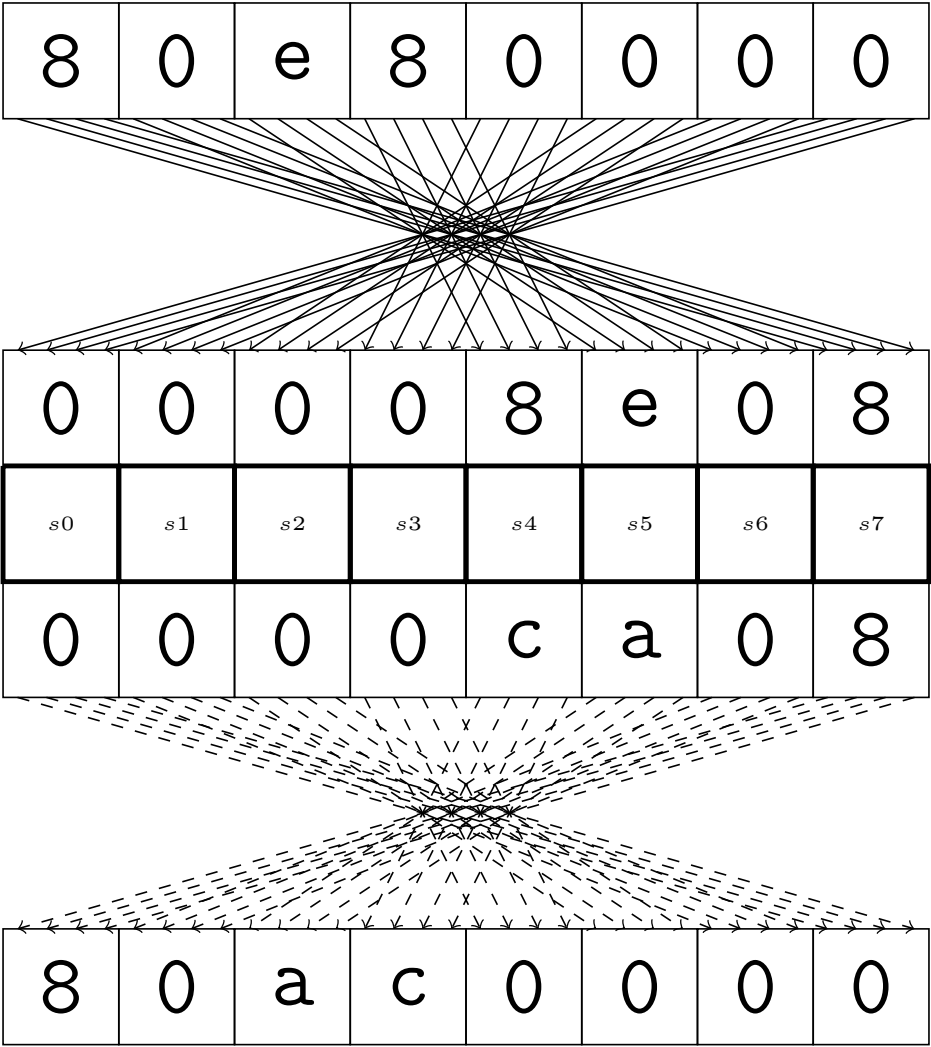


12 LBlock_round

Maximal linear correlation: $2^{-3.0}$ 

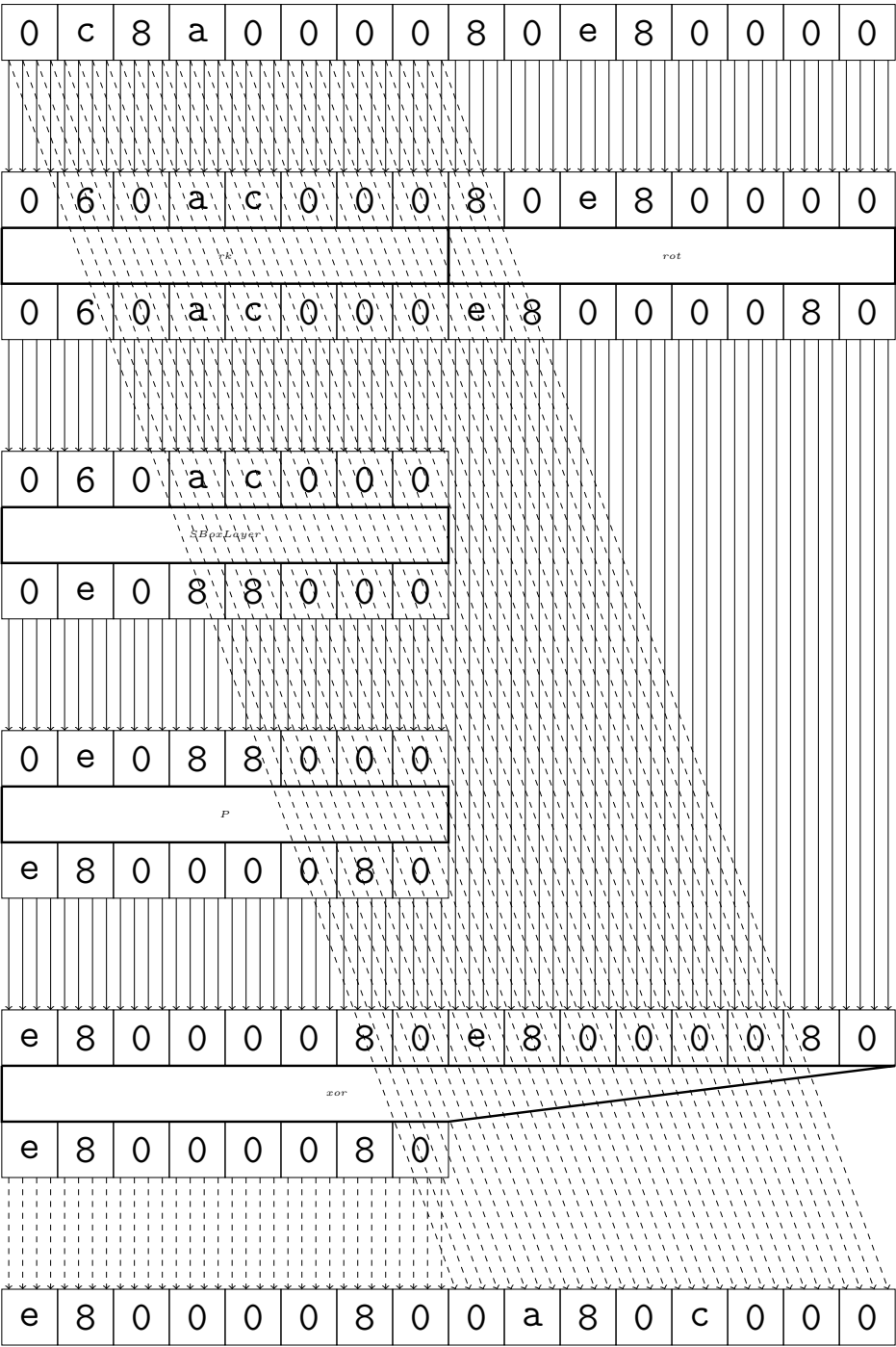
13 SBoxLayer

Maximal linear correlation: $2^{-3.0}$



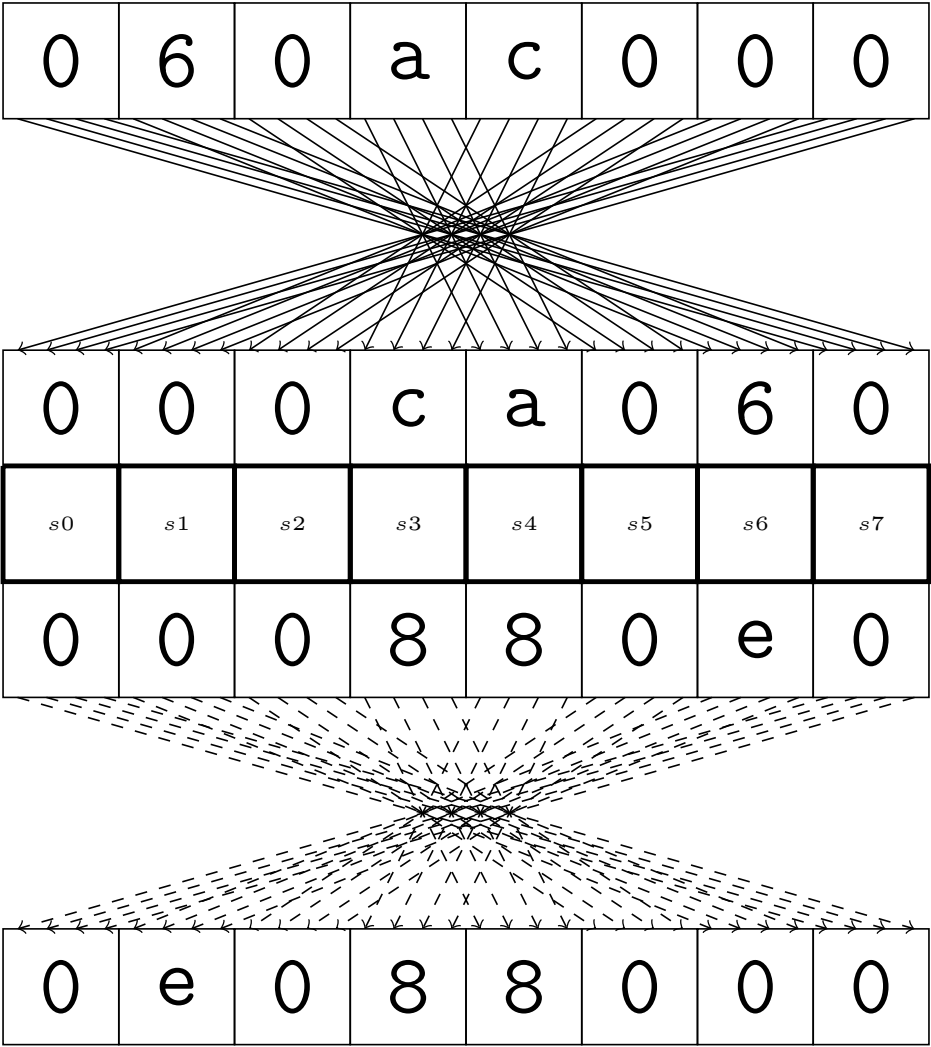
14 LBlock_round

Maximal linear correlation: $2^{-3.0}$



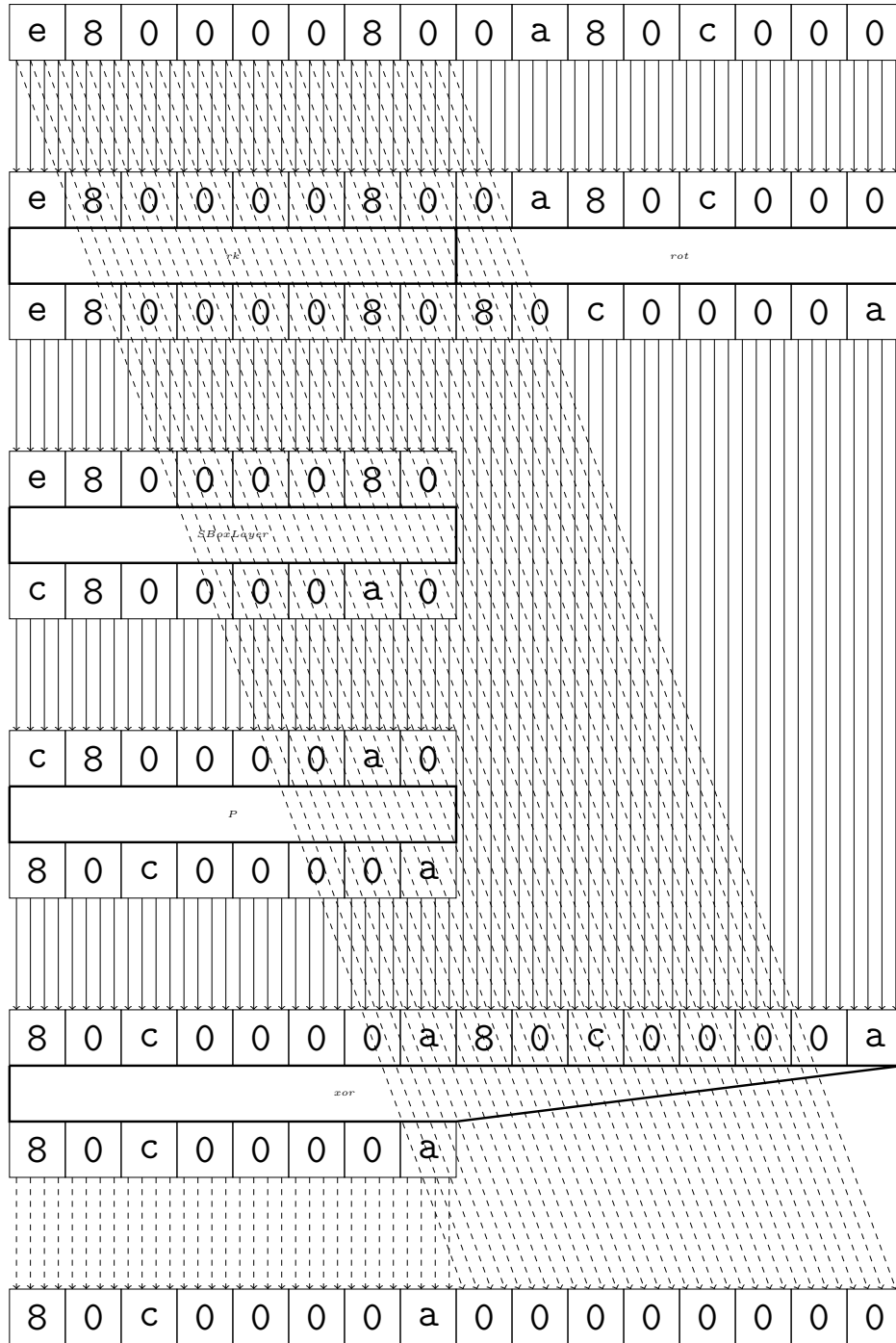
15 SBoxLayer

Maximal linear correlation: $2^{-3.0}$



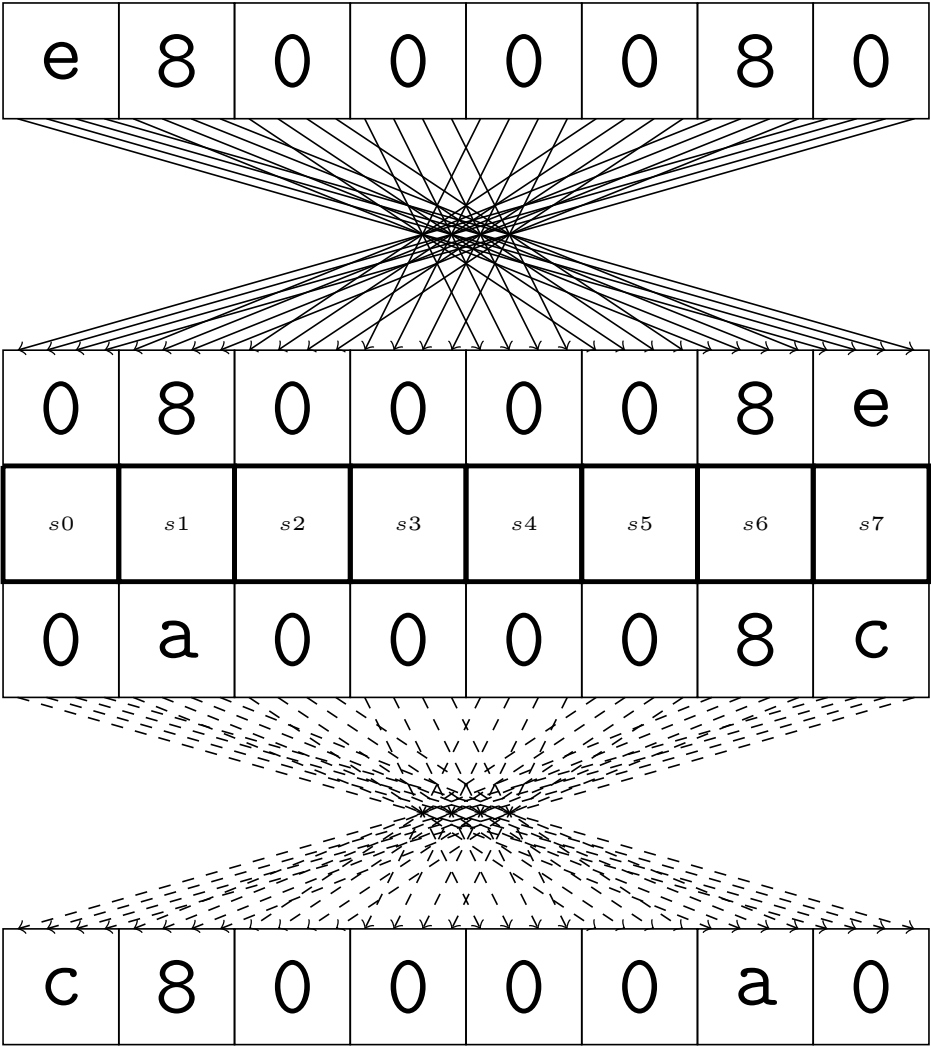
16 LBlock_round

Maximal linear correlation: $2^{-3.0}$



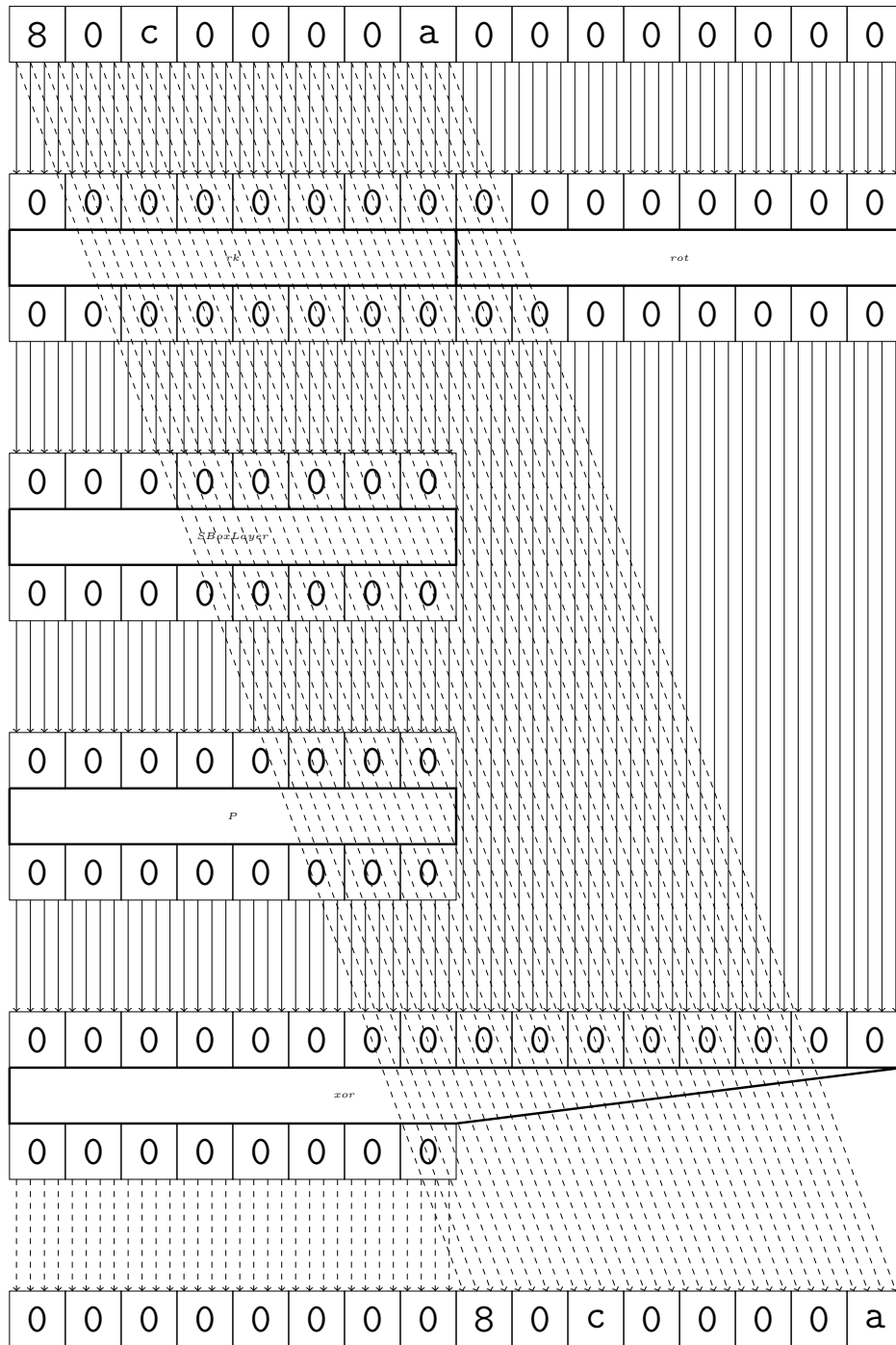
17 SBoxLayer

Maximal linear correlation: $2^{-3.0}$



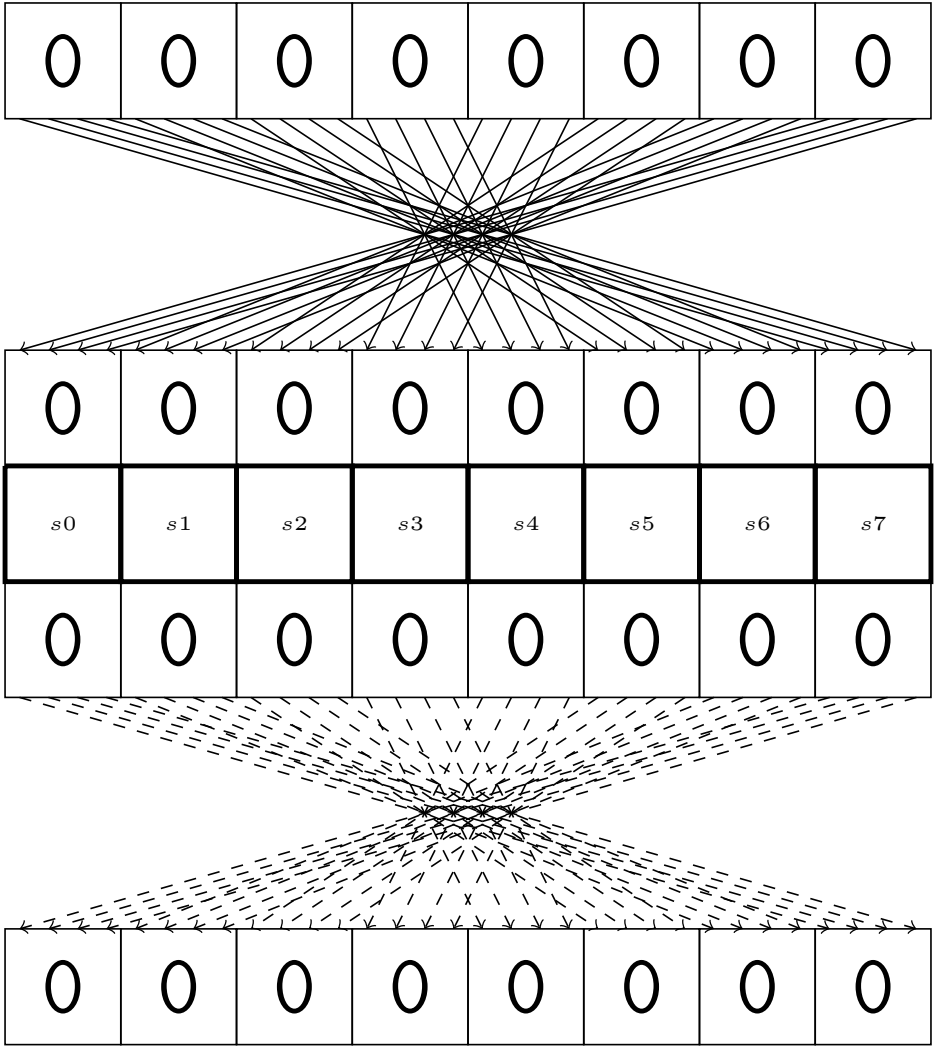
18 LBlock_round

Maximal linear correlation: $2^{-0.0}$



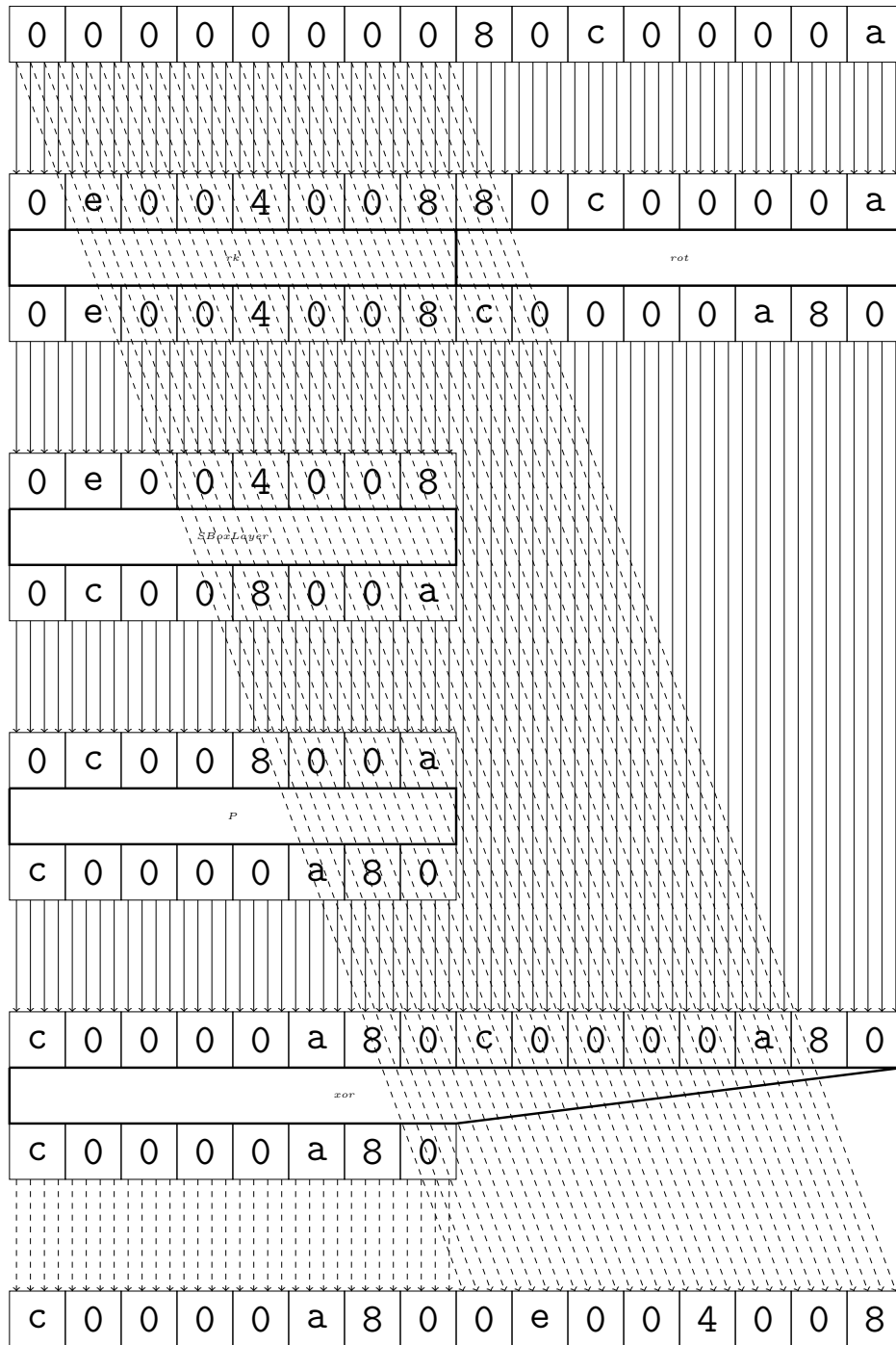
19 SBoxLayer

Maximal linear correlation: $2^{-0.0}$



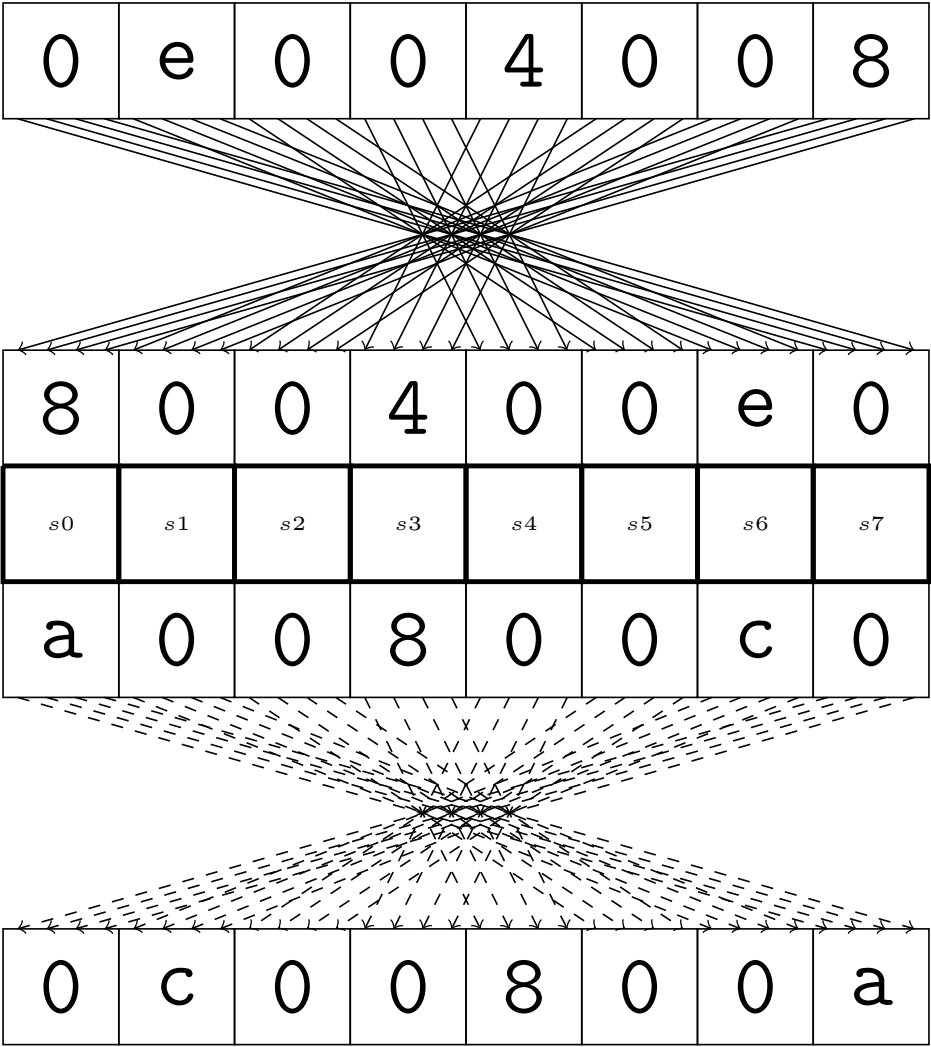
20 LBlock_round

Maximal linear correlation: $2^{-3.0}$

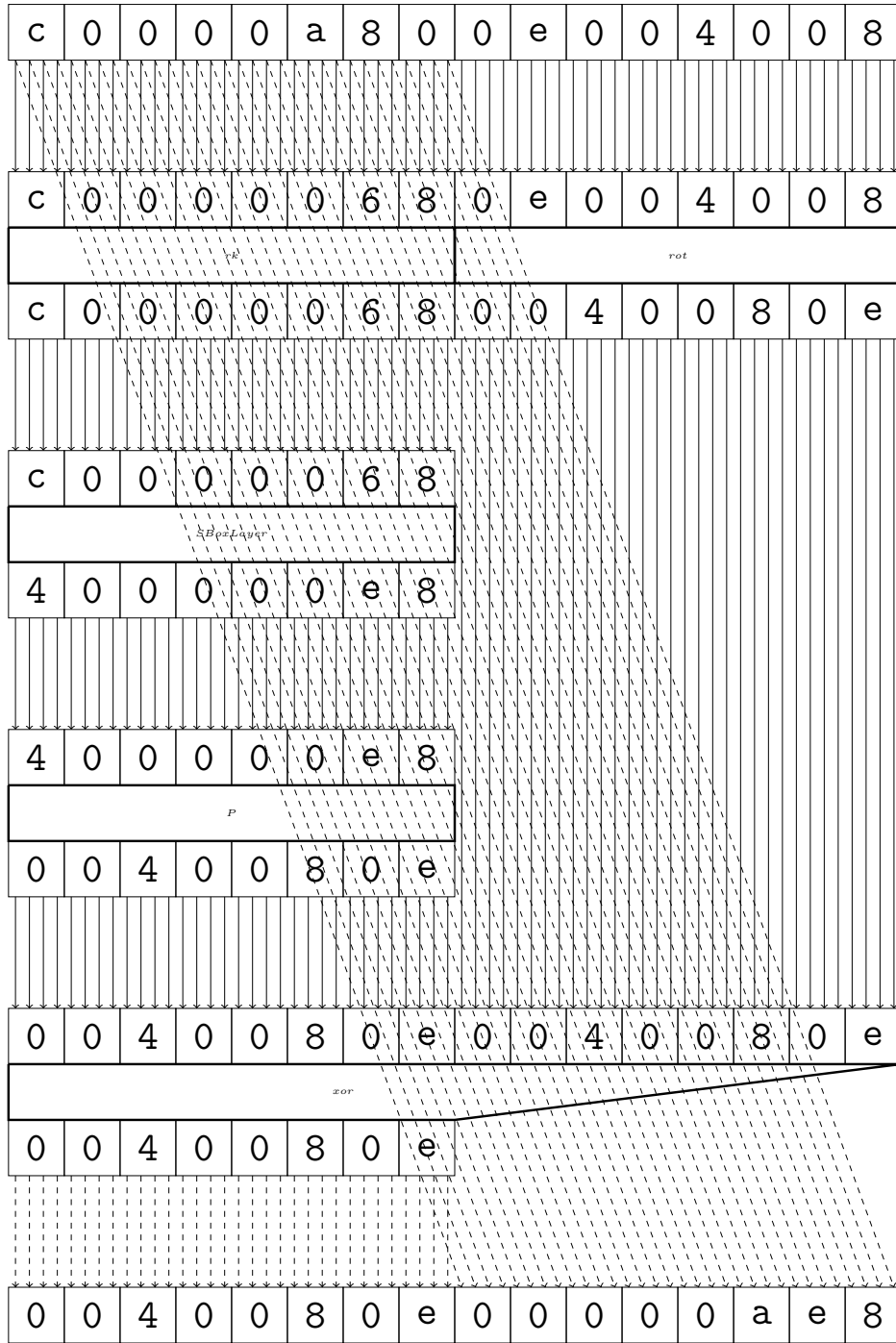


21 SBoxLayer

Maximal linear correlation: $2^{-3.0}$

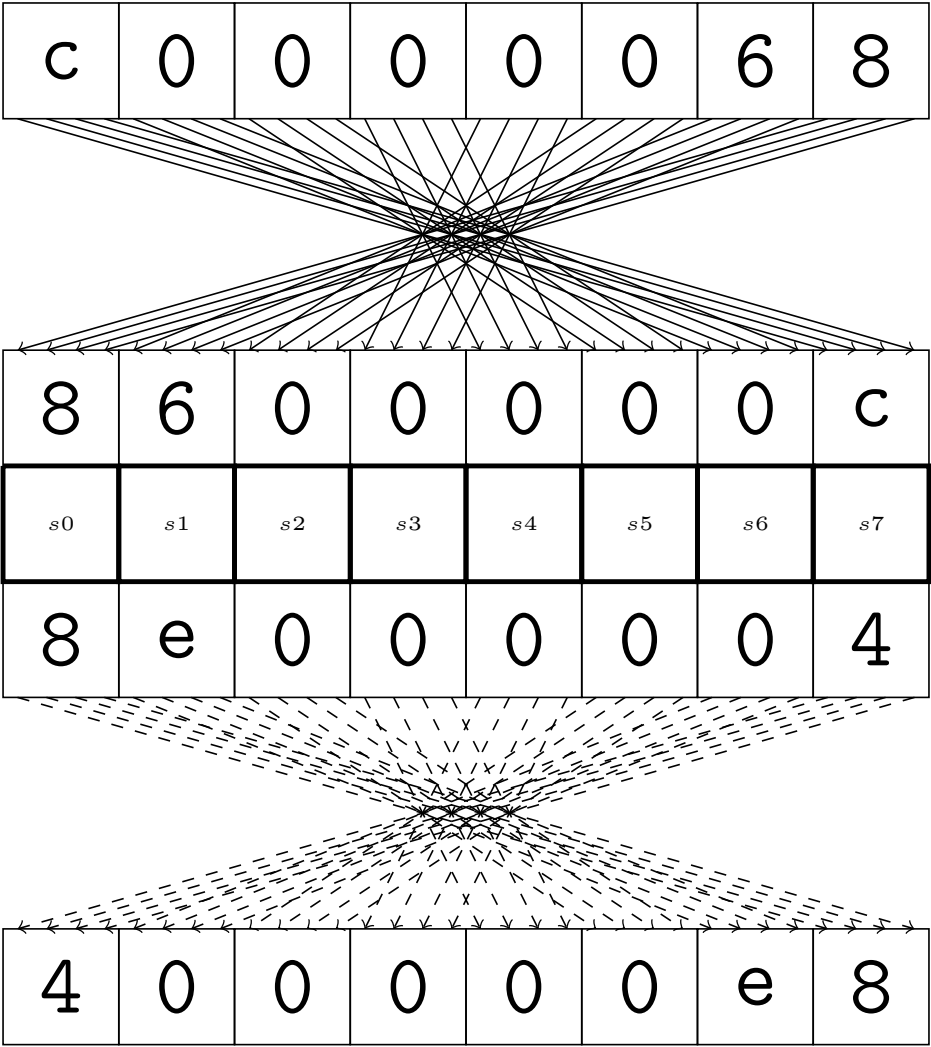


22 LBlock_round

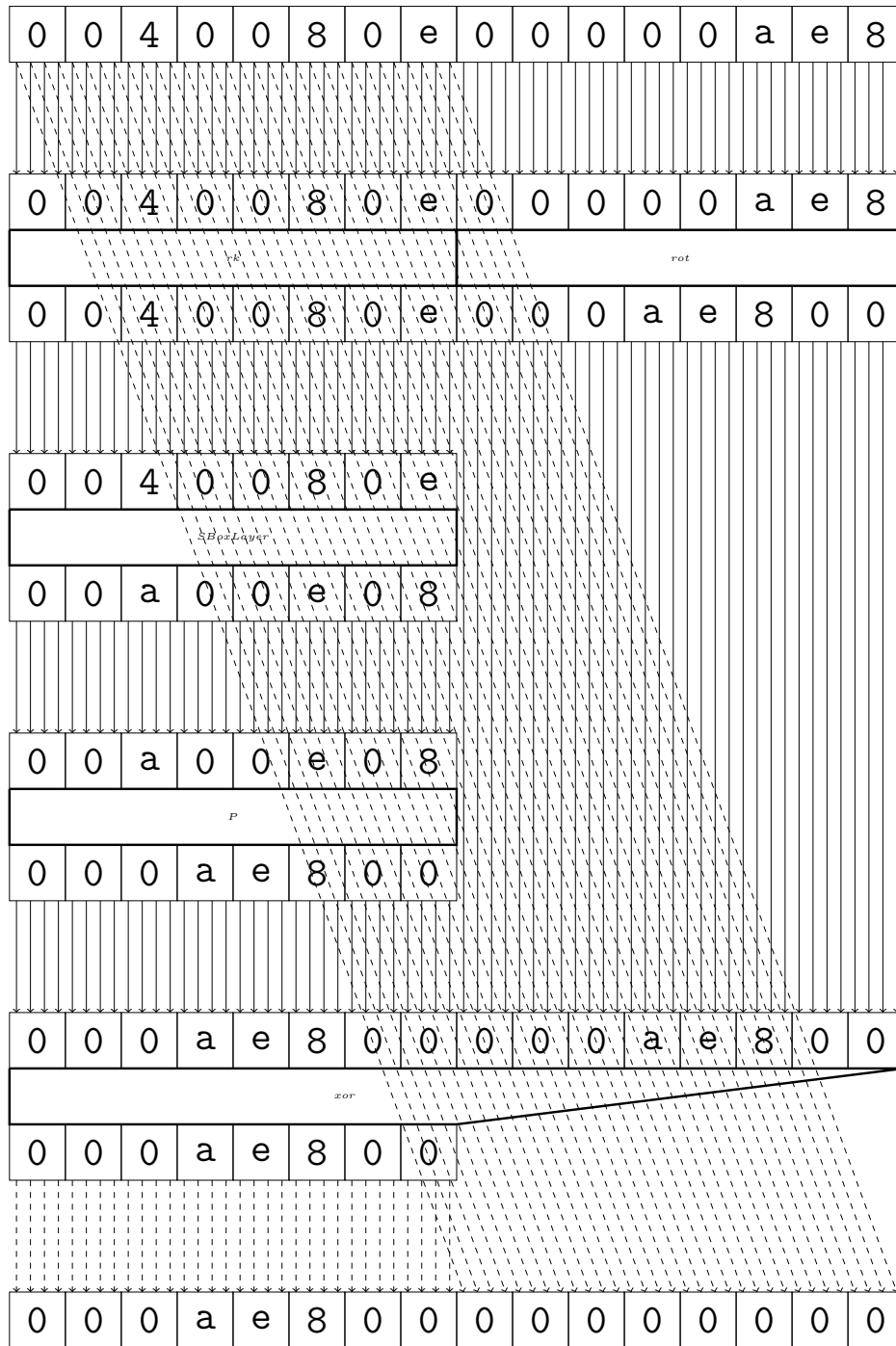
Maximal linear correlation: $2^{-3.0}$ 

23 SBoxLayer

Maximal linear correlation: $2^{-3.0}$

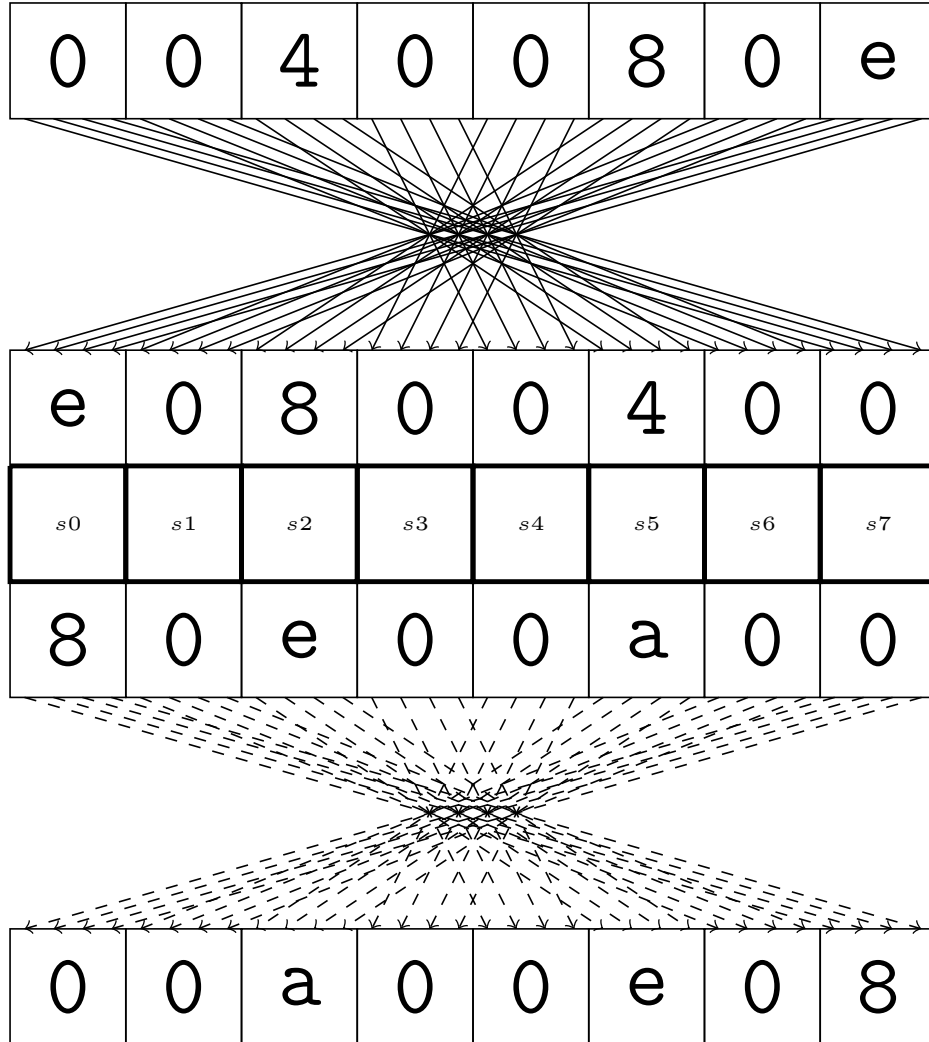


24 LBlock_round

Maximal linear correlation: $2^{-4.0}$ 

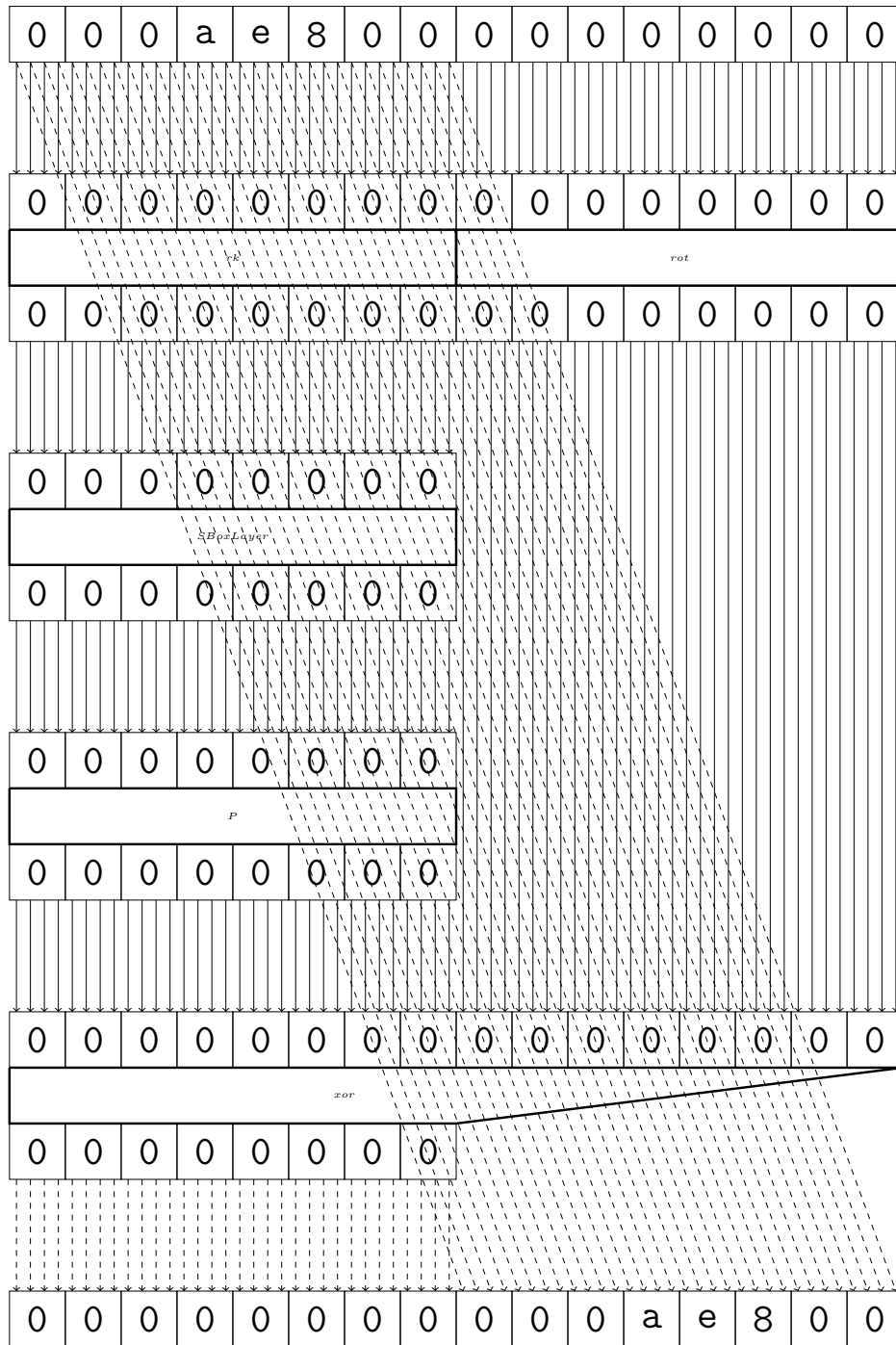
25 SBoxLayer

Maximal linear correlation: $2^{-4.0}$



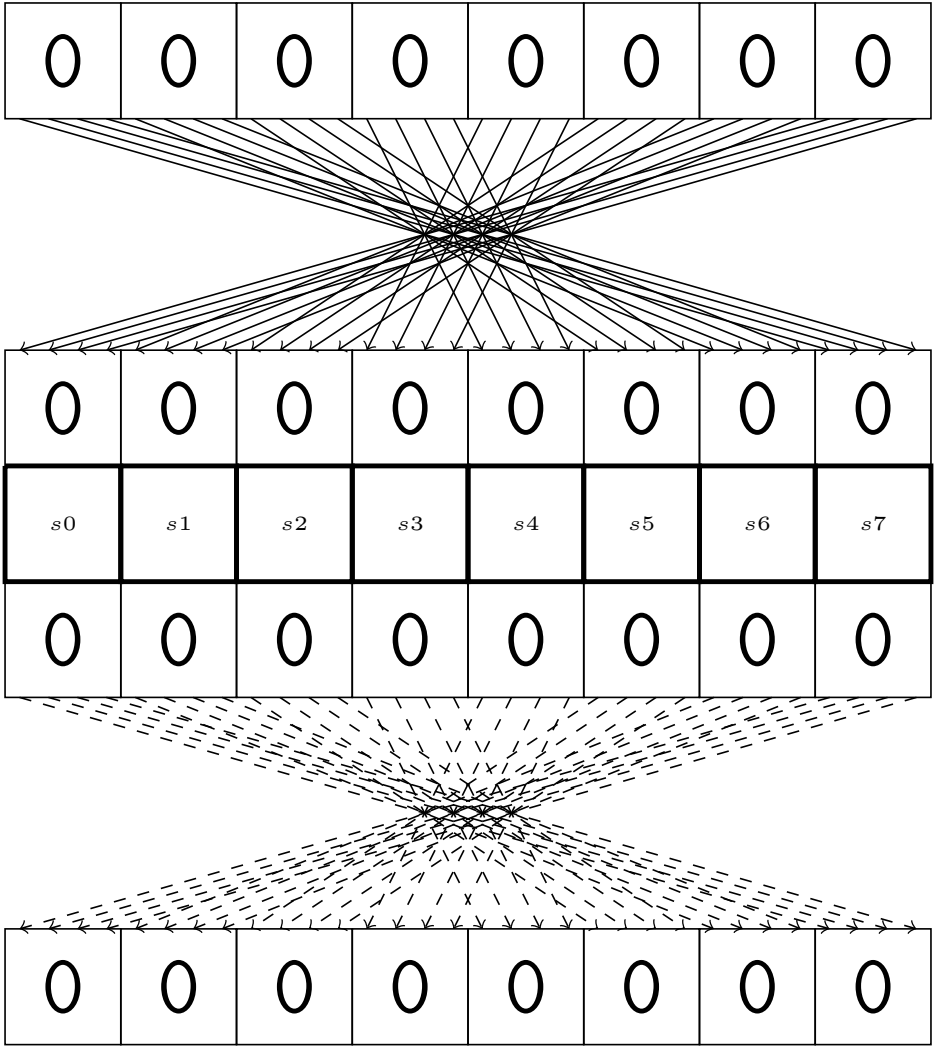
26 LBlock_round

Maximal linear correlation: $2^{-0.0}$



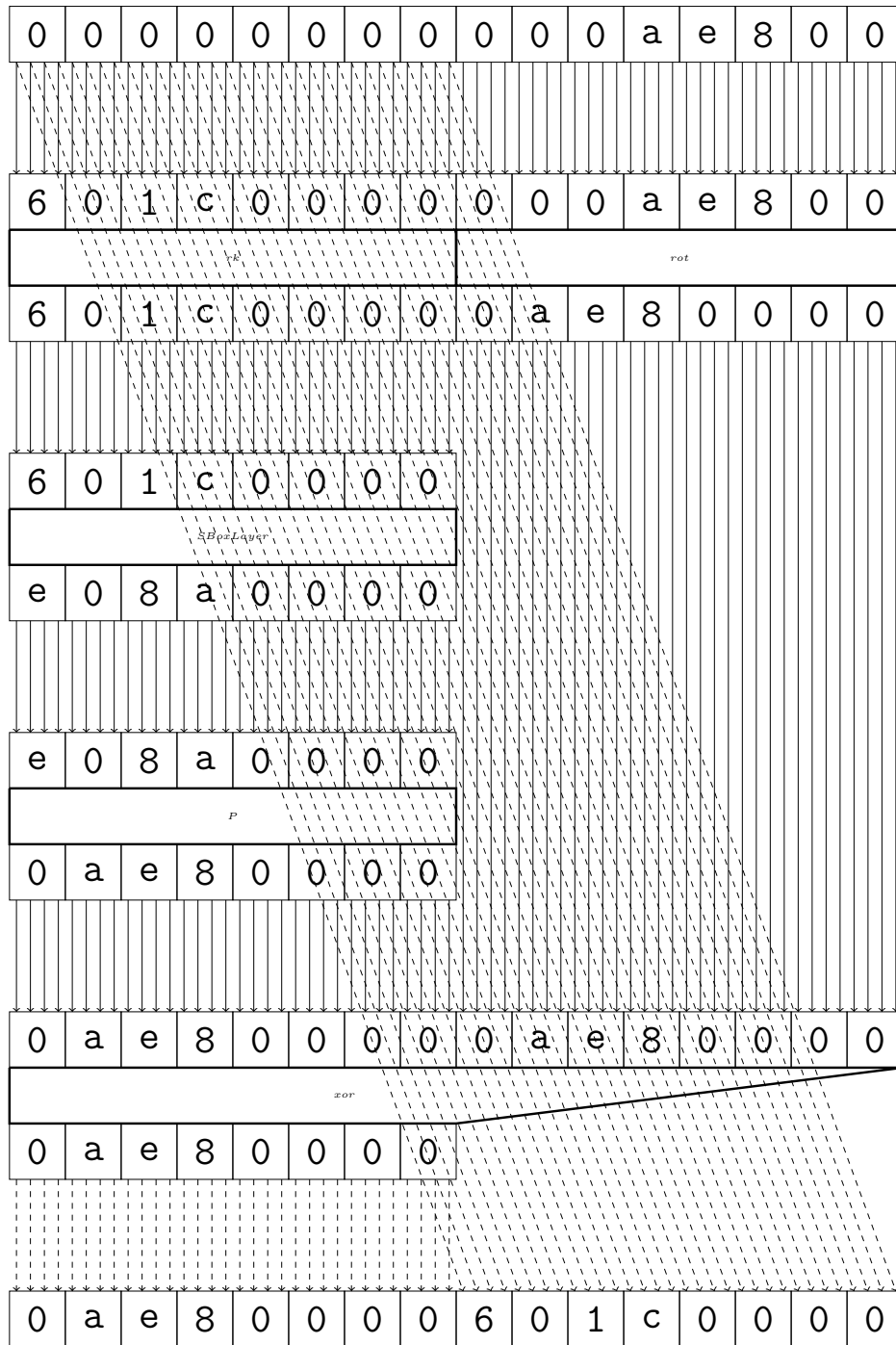
27 SBoxLayer

Maximal linear correlation: $2^{-0.0}$



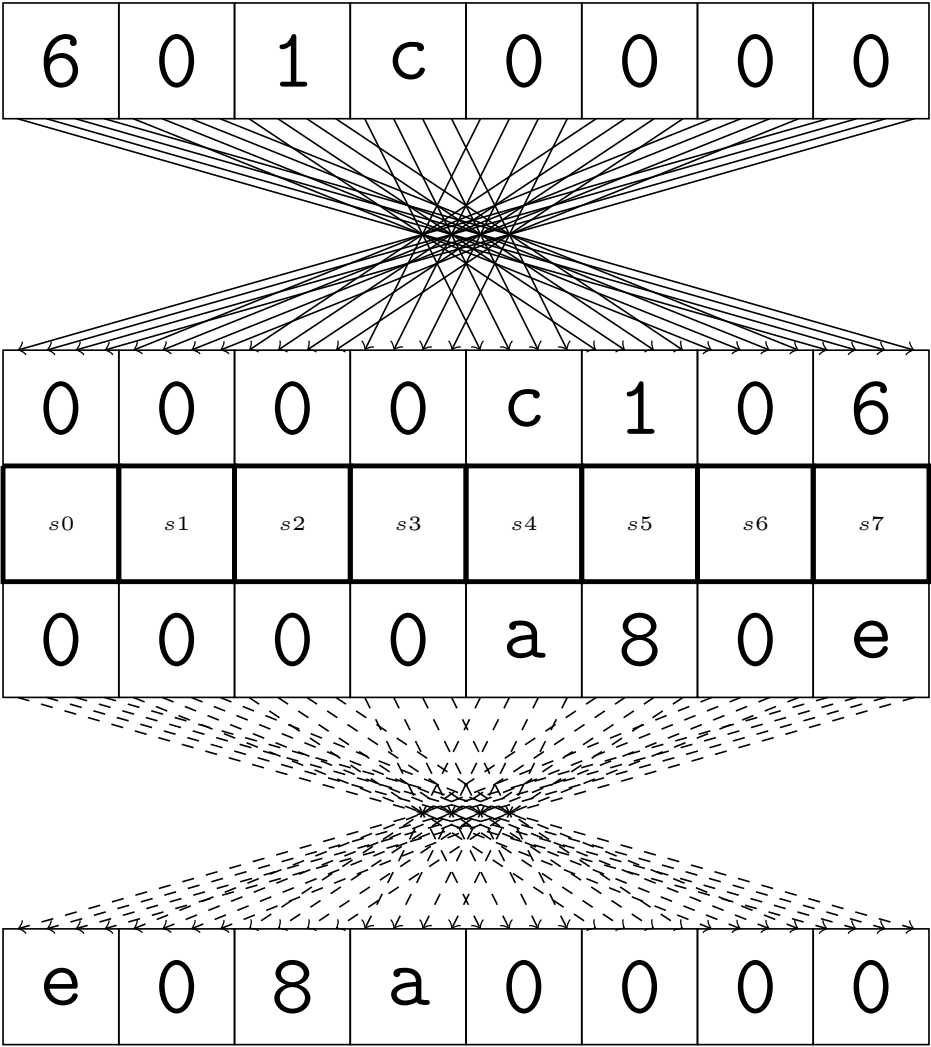
28 LBlock_round

Maximal linear correlation: $2^{-3.0}$

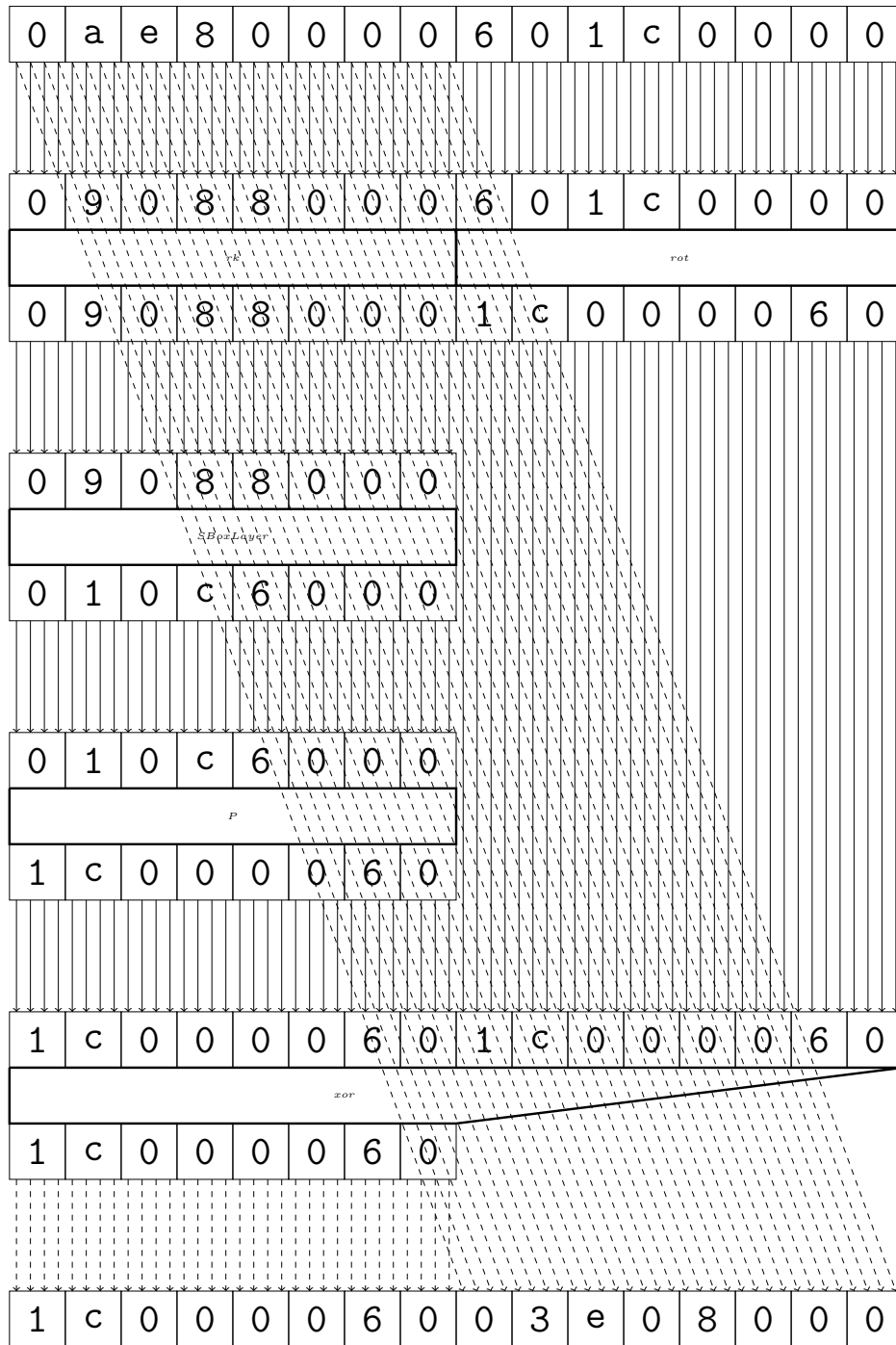


29 SBoxLayer

Maximal linear correlation: $2^{-3.0}$

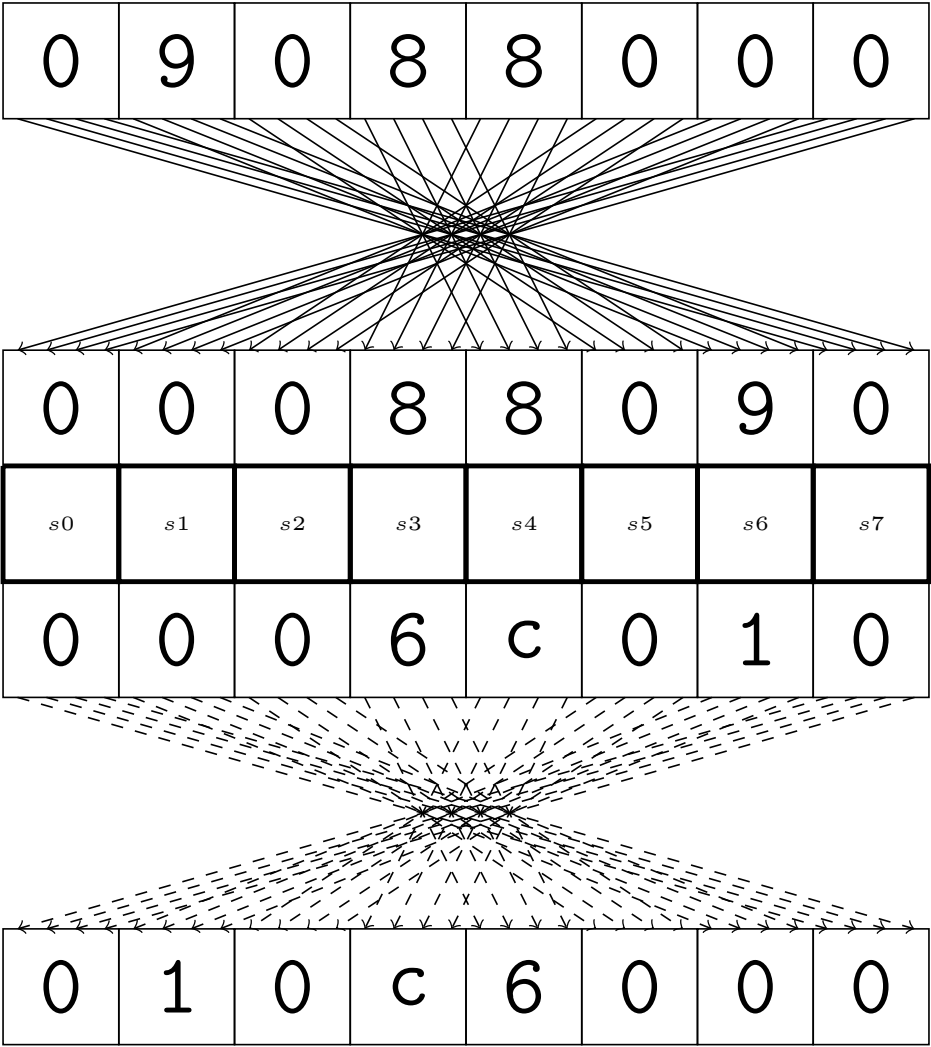


30 LBlock_round

Maximal linear correlation: $2^{-3.0}$ 

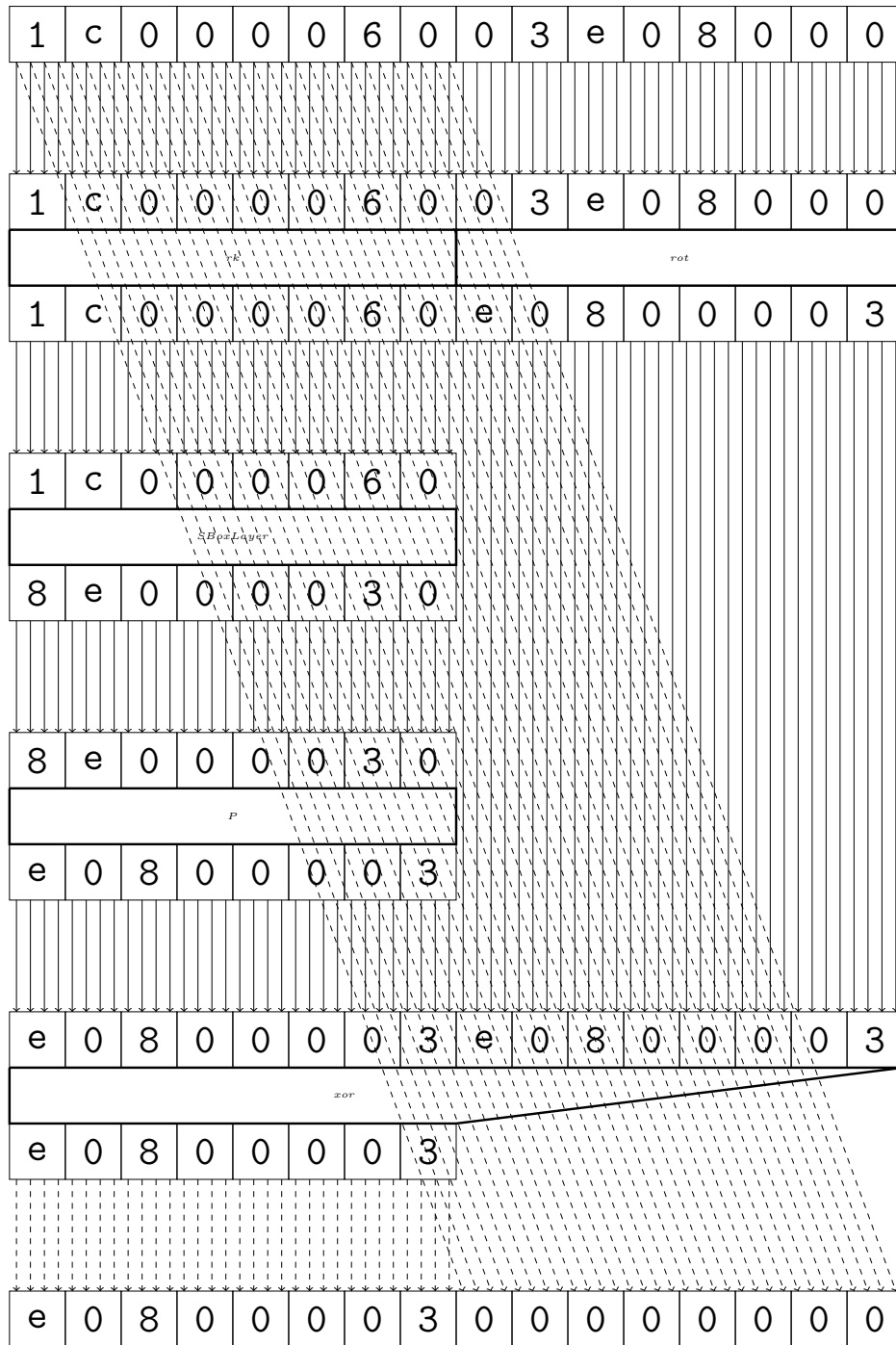
31 SBoxLayer

Maximal linear correlation: $2^{-3.0}$



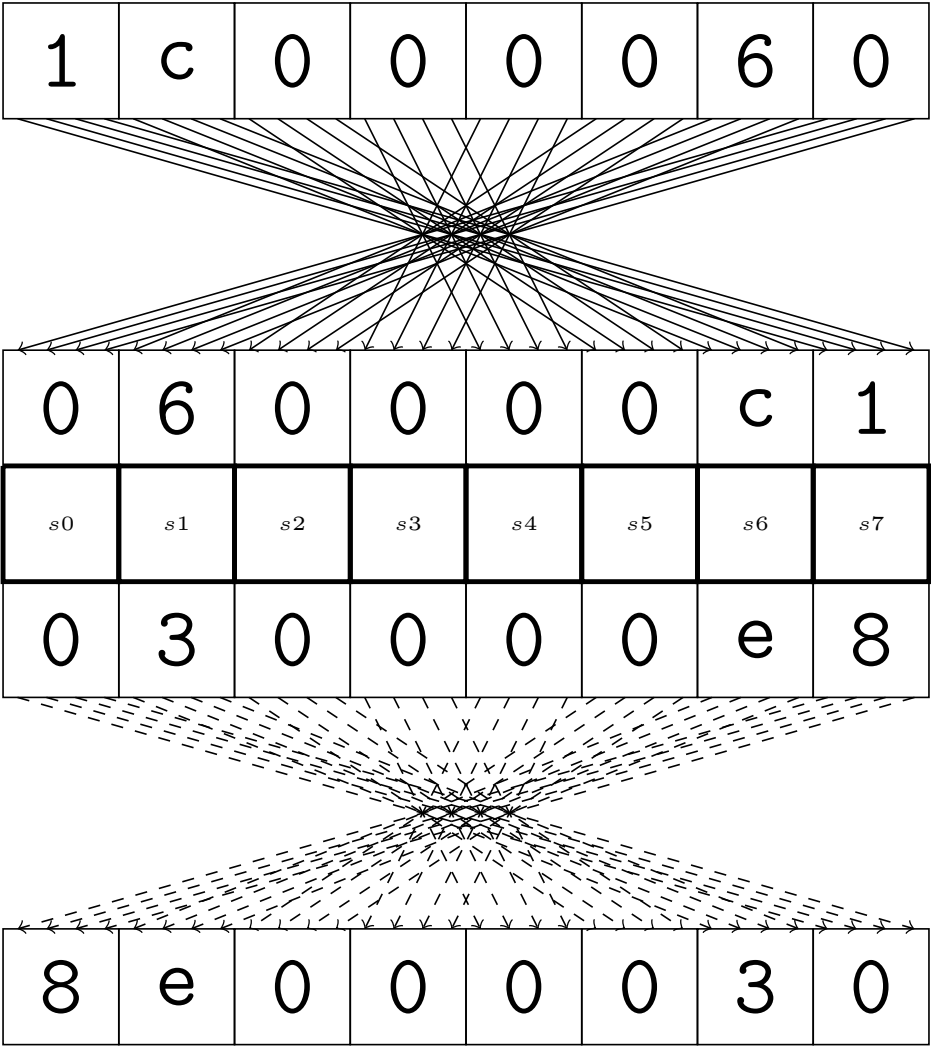
32 LBlock_round

Maximal linear correlation: $2^{-3.0}$



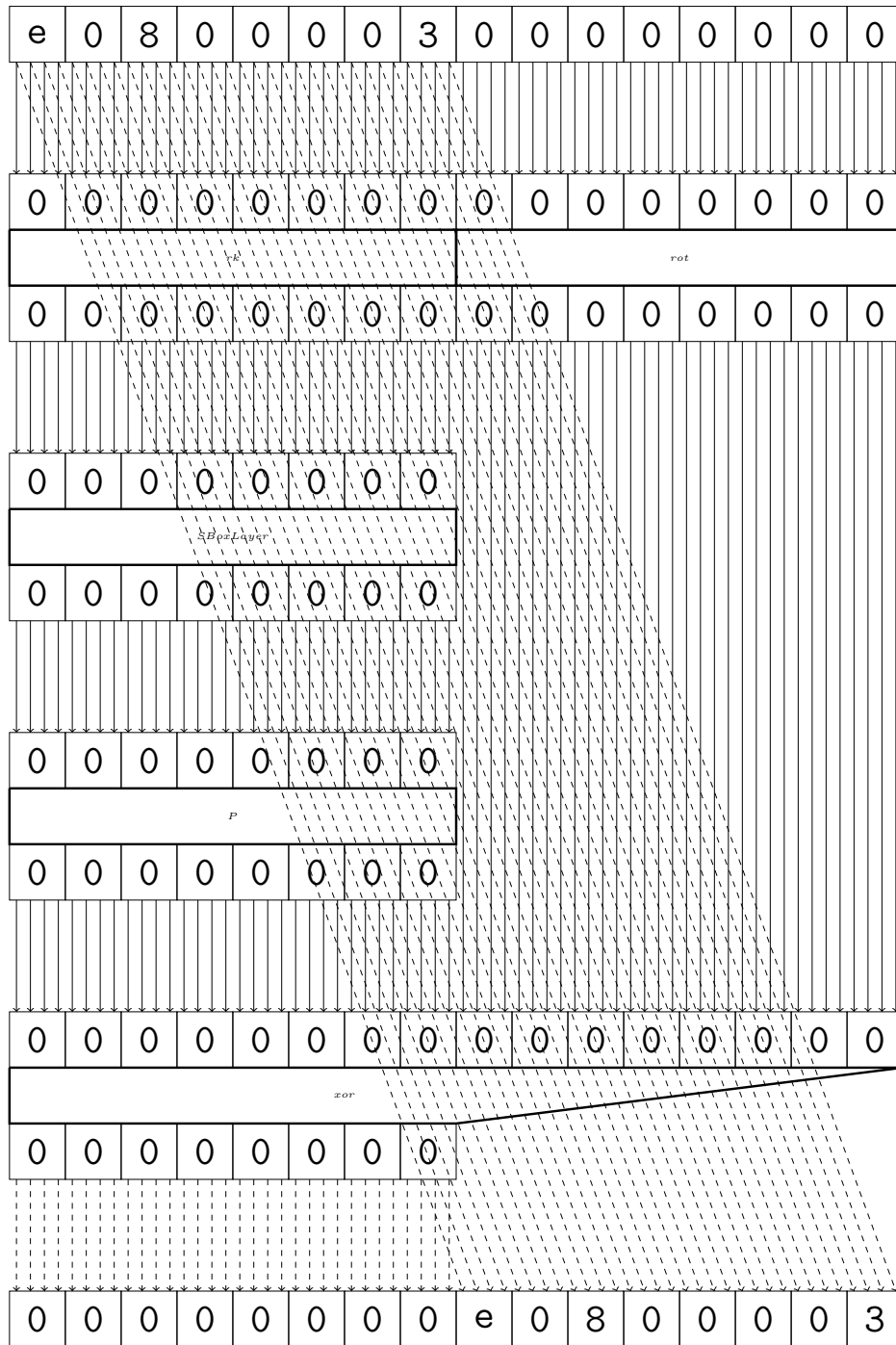
33 SBoxLayer

Maximal linear correlation: $2^{-3.0}$



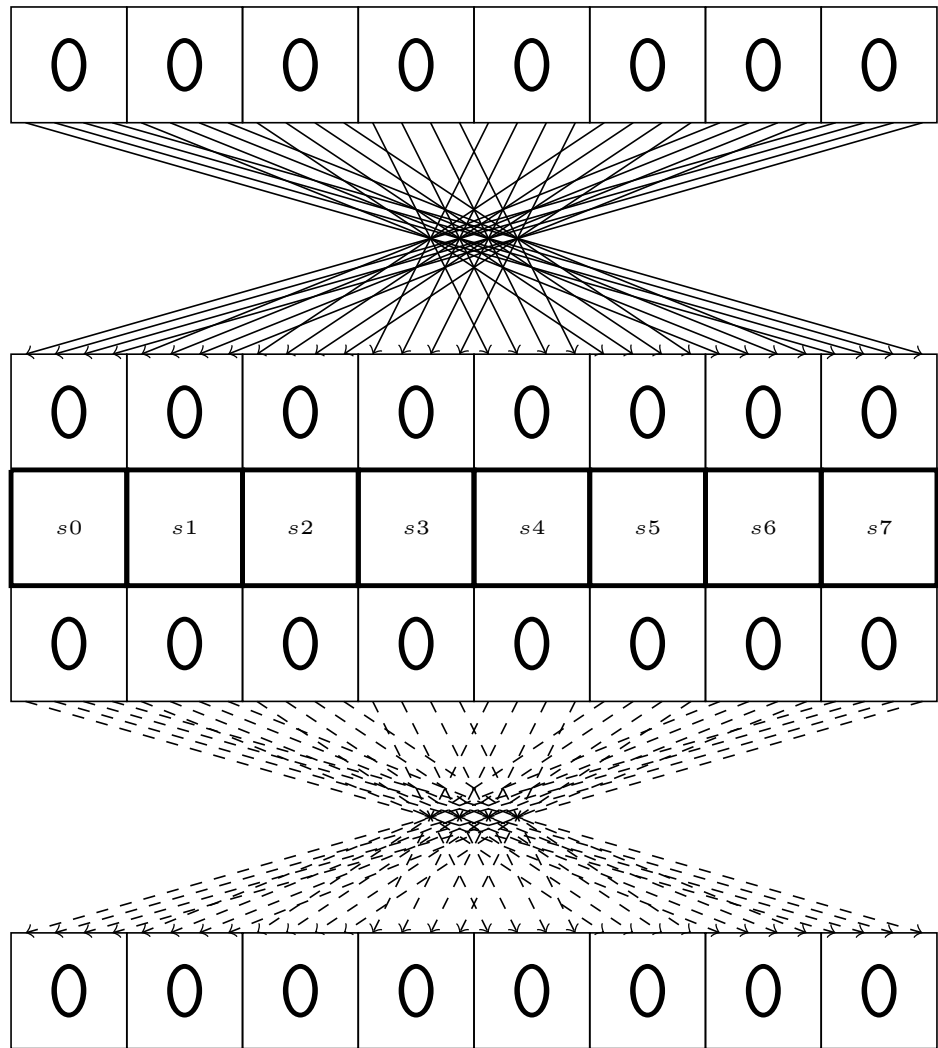
34 LBlock_round

Maximal linear correlation: $2^{-0.0}$



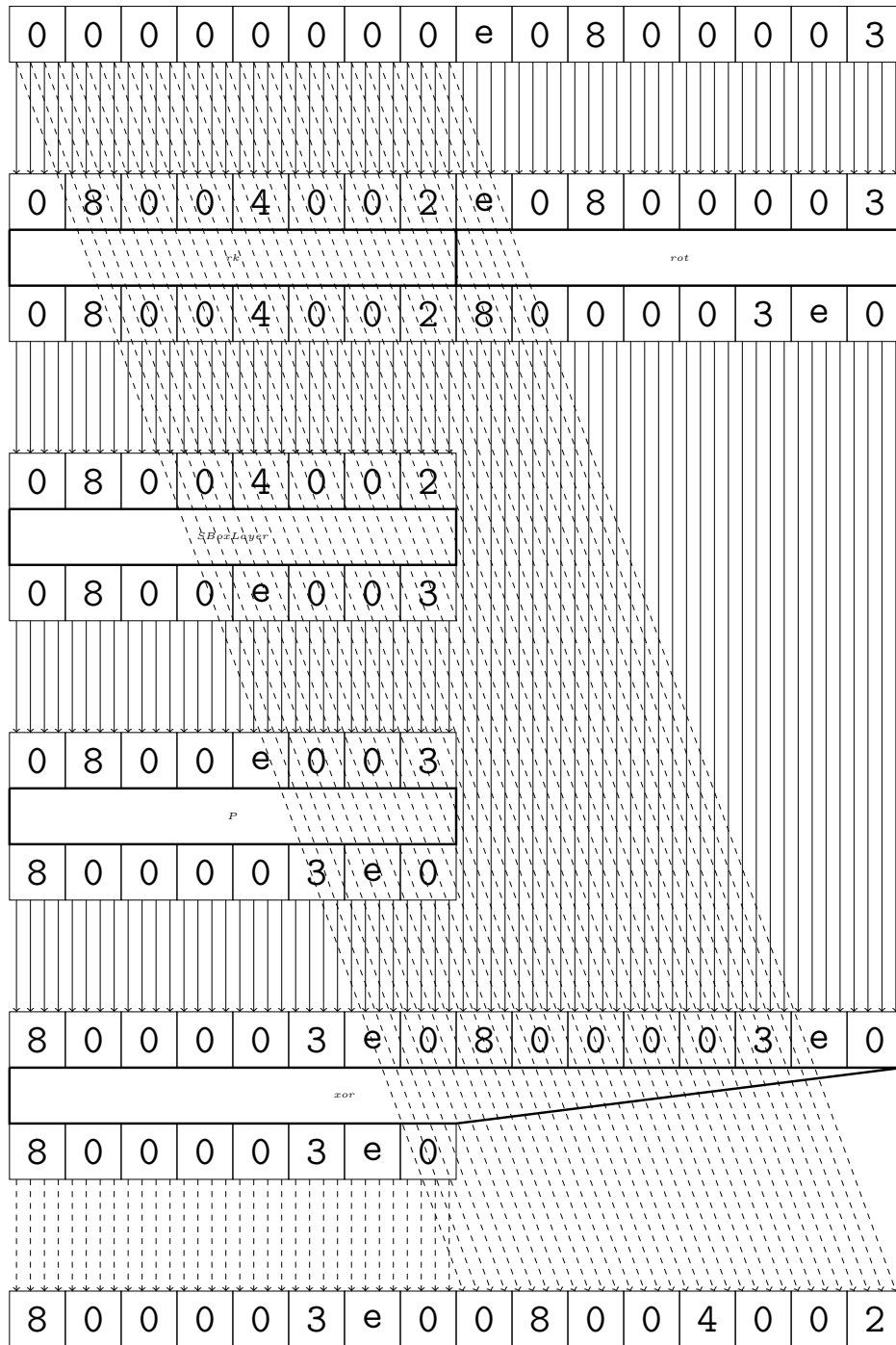
35 SBoxLayer

Maximal linear correlation: $2^{-0.0}$



36 LBlock_round

Maximal linear correlation: $2^{-3.0}$



37 SBoxLayer

Maximal linear correlation: $2^{-3.0}$

